Trilobite evolutionary rates constrain the duration of the Cambrian explosion

John R. Paterson[a,1], Gregory D. Edgecombe[b], and Michael S. Y. Lee[c,d]

[a]Palaeoscience Research Centre, School of Environmental & Rural Science, University of New England, Armidale, NSW 2351, Australia; [b]Department of Earth Sciences, The Natural History Museum, London SW7 5BD, United Kingdom; [c]College of Science and Engineering, Flinders University, SA 5001, Australia; and [d]Earth Sciences Section, South Australian Museum, Adelaide, SA 5000, Australia

Edited by Andrew H. Knoll, Harvard University, Cambridge, MA, and approved January 9, 2019 (received for review November 12, 2018)

Trilobites are often considered exemplary for understanding the Cambrian explosion of animal life, due to their unsurpassed diversity and abundance. These biomineralized arthropods appear abruptly in the fossil record with an established diversity, phylogenetic disparity, and provincialism at the beginning of Cambrian Series 2 (~521 Ma), suggesting a protracted but cryptic earlier history that possibly extends into the Precambrian. However, recent analyses indicate elevated rates of phenotypic and genomic evolution for arthropods during the early Cambrian, thereby shortening the phylogenetic fuse. Furthermore, comparatively little research has been devoted to understanding the duration of the Cambrian explosion, after which normal Phanerozoic evolutionary rates were established. We test these hypotheses by applying Bayesian tip-dating methods to a comprehensive dataset of Cambrian trilobites. We show that trilobites have a Cambrian origin, as supported by the trace fossil record and molecular clocks. Surprisingly, they exhibit constant evolutionary rates across the entire Cambrian, for all aspects of the preserved phenotype: discrete, meristic, and continuous morphological traits. Our data therefore provide robust, quantitative evidence that by the time the typical Cambrian fossil record begins (~521 Ma), the Cambrian explosion had already largely concluded. This suggests that a modern-style marine biosphere had rapidly emerged during the latest Ediacaran and earliest Cambrian (~20 million years), followed by broad-scale evolutionary stasis throughout the remainder of the Cambrian.

Cambrian explosion | evolutionary rates | trilobites | Bayesian tip-dating | morphological clock

The abrupt first appearance of a multitude of animal fossils in early Cambrian rocks (Terreneuvian to Series 2; ca. 541–509 Ma) epitomizes one of the most significant evolutionary events in Earth's history (1). This sudden burst of diversity and abundance across most eumetazoan (especially bilaterian) phyla over a relatively short geologic time span, and lack of obvious Precambrian precursors, poses a conundrum when attempting to reconcile the fossil record with the true tempo of early animal evolution. This issue even troubled Darwin (2) because it challenged his ideas on gradual evolutionary change. He suggested that the incompleteness of the geologic record can account for a protracted, cryptic history of animals before their appearance as diverse fossils. Over the 150+ years since *On the Origin of Species* was published, fossil discoveries in Ediacaran and Cambrian rocks and advances in chronostratigraphy, geochronology, and molecular clocks have diminished Darwin's dilemma (3, 4). However, there remain conspicuous gaps in the Cambrian records of many animal lineages—for example, the decoupled first appearances of euarthropod trace and body fossils (5)—perpetuating the idea of an older hidden history for many clades.

Fast evolutionary rates during the early Cambrian have been used to explain the rapid emergence of animals, providing support for a more literal reading of the fossil record. Evidence consistent with the radiation of animals within a short time period (~20 Ma) includes radiometric ages that have refined the Cambrian timescale (e.g., ref. 6), as well as elevated rates of phenotypic and genomic evolution (7, 8). Rapid morphological and molecular evolution during the earliest Cambrian almost certainly underpinned the pronounced pulses of origination and diversification throughout the Terreneuvian (3, 9, 10). However, the question remains as to when evolutionary rates slowed to Phanerozoic norms, thus marking the end of the Cambrian explosion. For instance, the calibrations used in ref. 7 were mostly 488 Ma or younger; that analysis therefore only had weak power to constrain fast early rates further back than that time point. Indirect measures using trends in animal diversity and disparity suggest that rates were elevated throughout the early Cambrian (3, 9, 10), but no study has yet quantified rates of evolution across a broad selection of Cambrian lineages using direct phenotypic information from the fossil record.

Trilobites are a diverse and abundant clade of biomineralized crown-group euarthropods that best exemplify the disjunct between the Cambrian rock record and any expected gradualist history of a clade before its first appearance as fossils. The oldest trilobite body fossils around the world, at or near the Terreneuvian–Cambrian Series 2 boundary (ca. 521 Ma), already show established diversity, phylogenetic disparity, and biogeographic provincialism (11–13). This, among other evidence, has been used to suggest that trilobites had a much earlier, Precambrian origin (e.g., refs. 14–16). In fact, Darwin (2) chose trilobites as an exemplar group to highlight his dilemma about animal origins: "There is another and allied difficulty, which is much graver. I allude to the manner in which

Significance

The Cambrian explosion was arguably the most important biological event after the origin of life. Extensive research has been devoted to understanding when it began but far less on when this burst of evolution ended. We present a quantitative study that addresses these issues, using a large new dataset of Cambrian trilobites, the most abundant and diverse organisms during this time. Using probabilistic clock methods, we calculate rates of evolution in the earliest trilobites virtually identical to those throughout their Cambrian fossil history. We conclude that the Cambrian explosion was over by the time the typical Cambrian fossil record commences and reject an unfossilized Precambrian history for trilobites, solving a problem that had long troubled biologists since Darwin.

Author contributions: J.R.P., G.D.E., and M.S.Y.L. designed research; J.R.P., G.D.E., and M.S.Y.L. performed research; M.S.Y.L. analyzed data; J.R.P. and G.D.E. collected phenotypic and stratigraphic data; and J.R.P., G.D.E., and M.S.Y.L. wrote the paper.

The authors declare no conflict of interest.

This article is a PNAS Direct Submission.

Published under the PNAS license.

Data deposition: Data related to this work has been deposited in the Dryad Digital Repository (doi:10.5061/dryad.v7q827k).

[1]To whom correspondence should be addressed. Email: jpater20@une.edu.au.

This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1819366116/-/DCSupplemental.

Published online February 19, 2019.

numbers of species of the same group, suddenly appear in the lowest known fossiliferous rocks... For instance, I cannot doubt that all the [Cambrian] trilobites have descended from some one crustacean, which must have lived long before the [Cambrian] age" (p. 306).

Here we test Darwin's hypothesis (2) and later claims of elevated evolutionary rates during the early Cambrian (e.g., refs. 6 and 7) by analyzing an extensive dataset of Cambrian trilobites using Bayesian tip-dating clock methods (17). The phylogenetic dataset is the largest and most comprehensive for trilobites compiled to date, comprising 107 species—representing most Cambrian families (sensu ref. 18) that range from Series 2 to the Furongian (ca. 521–485 Ma)—and 115 traits that cover all aspects of the preserved phenotype [107 discrete, 2 meristic, and 6 continuous (*SI Appendix*, Fig. S1)]. To satisfy the methodology of tip-dating, this dataset explicitly sampled autapomorphies with the same intensity as cladistically informative traits. Where possible, species were preferentially selected based on fully articulated exoskeletons and known ontogenies. Stratigraphic ages for each species were determined by cross-referencing associated biozones with the calibrated Cambrian timescale (19) and other sources (*SI Appendix*).

Cambrian Evolutionary Rates

Phenotypic and stratigraphic data were analyzed using tip-dated Bayesian approaches (20, 21) that coestimate topologies, divergence dates, and evolutionary rates. To provide multiple independent estimates of evolutionary tempo across the Cambrian, rates of evolution of discrete, meristic, and continuous data were each estimated separately across time, using unlinked epoch clock models, which assume rates vary across time slices (but are shared across all lineages in the same time slice). Thus, rates of evolution for the 107 discrete characters were estimated for the early (Series

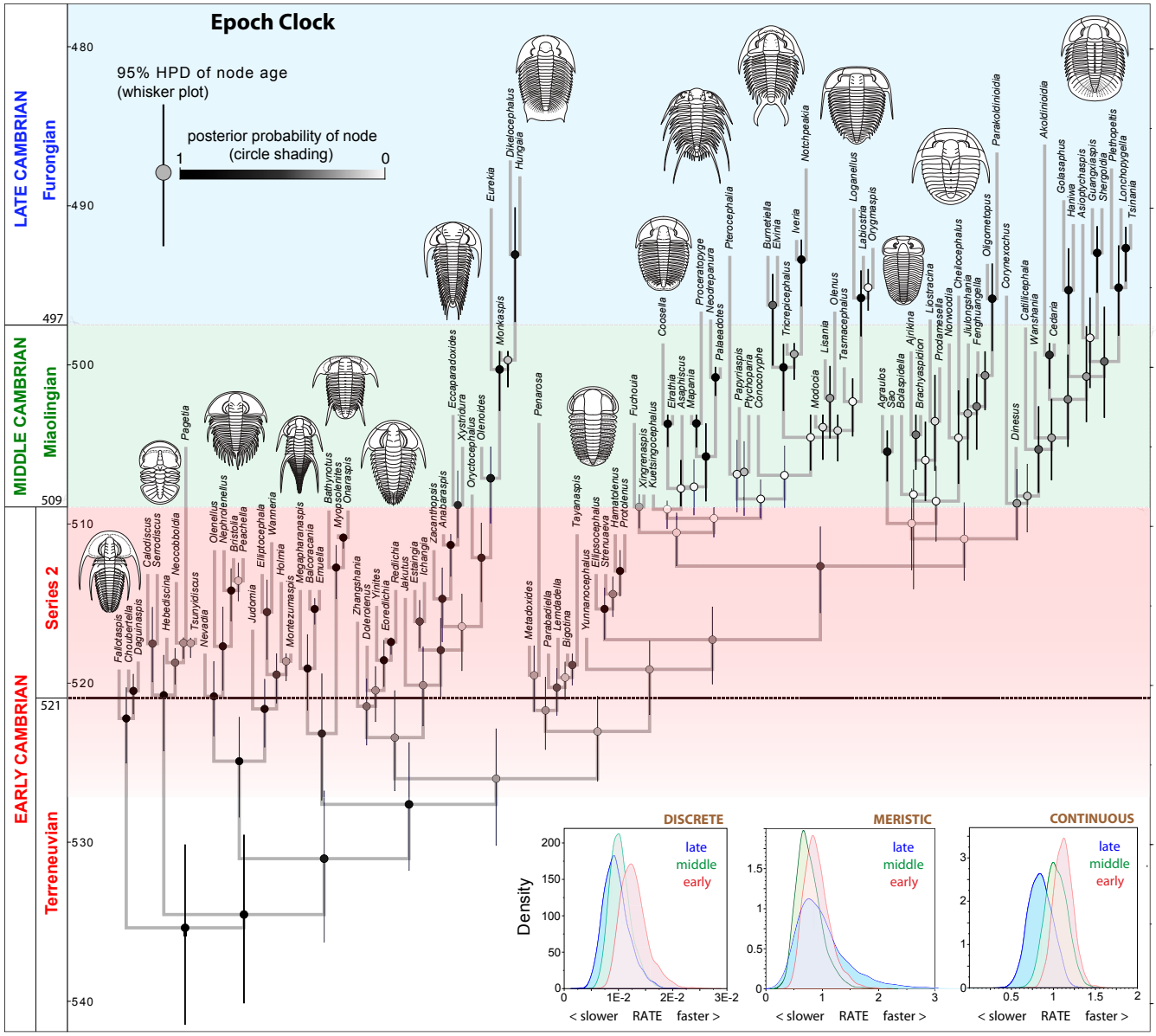


Fig. 1. Dated time tree of Cambrian trilobites inferred from tip-dated Bayesian analyses of discrete, meristic, and continuous traits under a multiepoch clock, which allows rates of evolution to vary across time slices. Evolutionary rates for discrete, meristic, and continuous traits were very constant across the early, middle, and late Cambrian. Notably, all three datasets failed to exhibit sharply elevated rates in the earliest time slice. Rate units are from raw BEAST (21) output; see *SI Appendix*, Table S1 for absolute and scaled rates. Full species names are presented in *SI Appendix*.

2, 521–509 Ma), middle (Miaolingian, 509–497 Ma), and late (Furongian, 497–485 Ma) Cambrian, and likewise (separately) for the two meristic and for the six continuous traits. Alternative models of evolutionary tempo were also evaluated using Bayes factors: a strict clock (which assumes rates are constant across time slices and across lineages) and an uncorrelated relaxed clock (which assumes rates vary across lineages but not necessarily systematically across time). Parsimony analyses were also performed to test the sensitivity of the tree topologies to analytical methods and to facilitate comparison of phylogeny inferred from phenotypic characters alone and those also incorporating temporal data. Phenotypic and stratigraphic data, details of all analyses, and executable scripts are in the *SI Appendix* and Dryad Digital Repository (doi.org/10.5061/dryad.v7q827k).

All analyses reveal that rates of morphological evolution were homogeneous throughout the Cambrian. In the epoch clock model, rates are marginally but insignificantly higher during the early Cambrian compared with the middle and late Cambrian (Fig. 1). Accordingly, a strict (or single-epoch) clock (*SI Appendix*, Fig. S2)—which assumes rates are homogeneous across the entire time period spanned by the sampled fossils—fits the data better than does the epoch clock model. The relaxed clock also returned very homogeneous rates of evolution across time (Figs. 2 and 3). This time-constant rate pattern is consistent across the discrete, meristic, and continuous characters: for all three trait types, rates of evolution are very uniform across the entire Cambrian trilobite fossil record. These results parallel the finding that speciation rates among certain Cambrian Series 2 trilobites were not

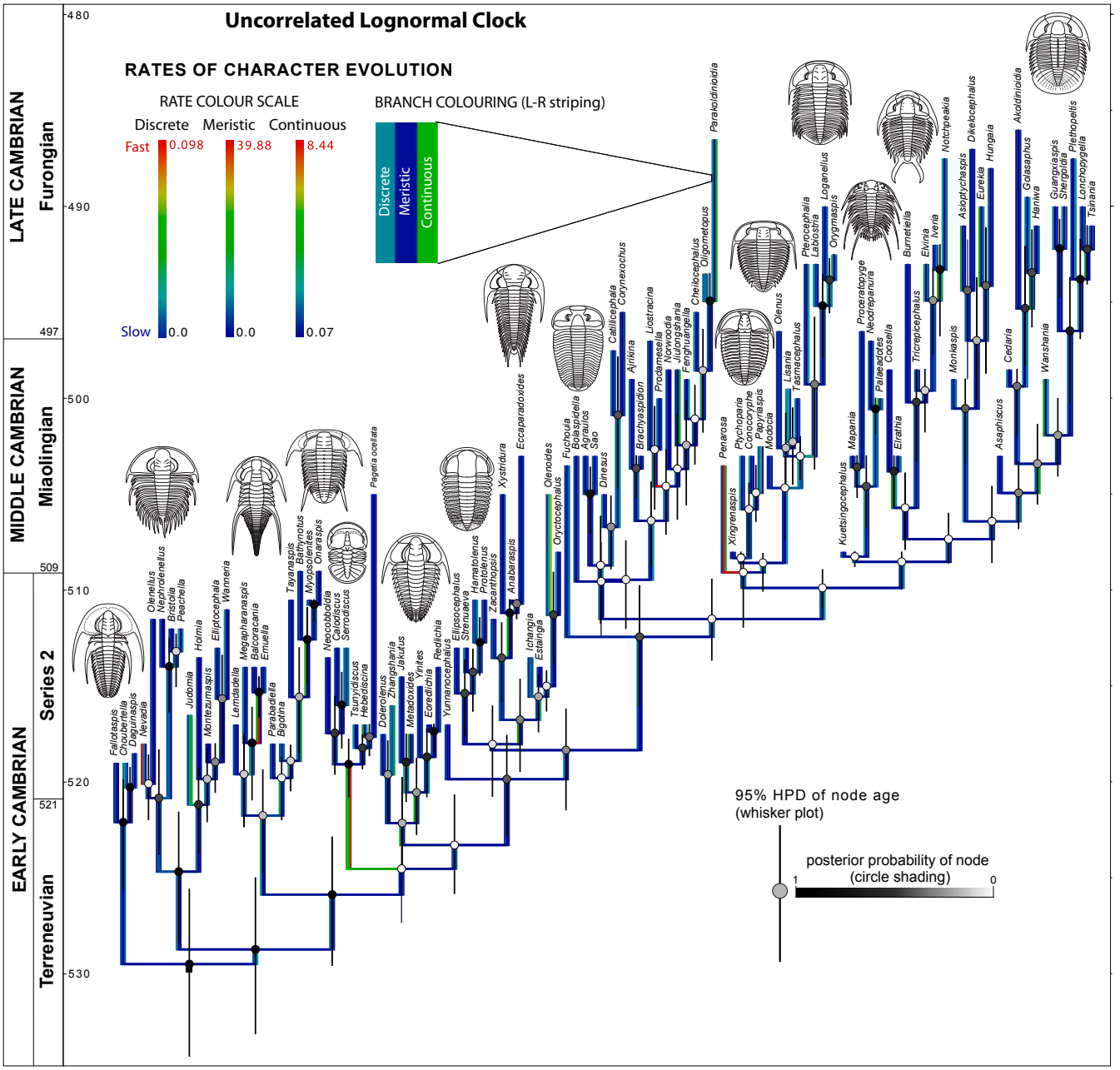


Fig. 2. Dated time tree of Cambrian trilobites inferred from tip-dated Bayesian analyses of discrete, meristic, and continuous traits under an uncorrelated lognormal (UCLN) relaxed clock, which allows rates of evolution to vary across all individual branches. Full species names are presented in the *SI Appendix*.

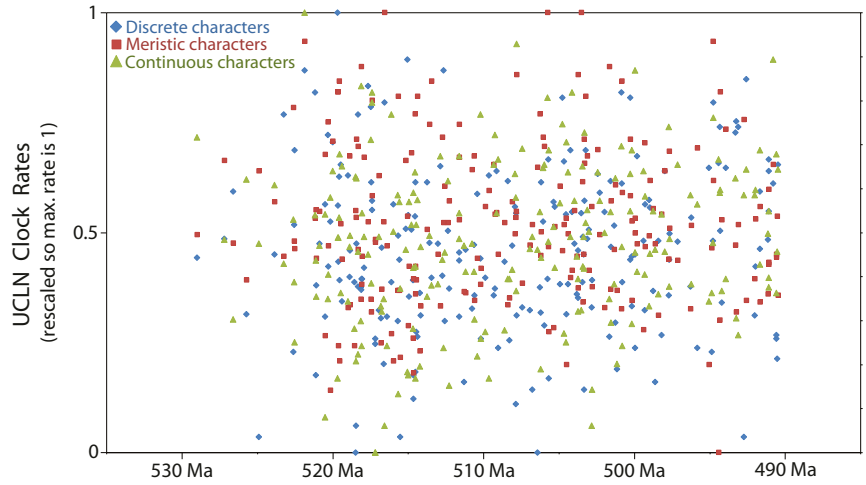


Fig. 3. Evolutionary rates for discrete, meristic, and continuous traits under an UCLN relaxed clock, showing that they were very constant across the early, middle, and late Cambrian. Rates have been rescaled so that the maximum rate is 1, to make the vertical axis comparable across discrete, meristic, and continuous characters.

unusually high, although slightly elevated relative to later times (22); however, there appears to be no obvious correlation between morphological evolutionary rates and levels of intraspecific morphological variation for Cambrian trilobites (23). Given the large number of taxa sampled (relative to the number of variable characters), there is substantial phylogenetic uncertainty, many nodes have weak support, and the consensus trees differ in certain clades between analyses (Figs. 1 and 2 and *SI Appendix*, Figs. S2 and S3). However, the above inferences of evolutionary rates accommodate this phylogenetic uncertainty by integrating all parameter estimates and error intervals across the full pool of sampled trees.

The basal divergence in Trilobita is estimated by our dated trees to be within the Terreneuvian (Figs. 1 and 2). In our epoch clock model (Fig. 1), this divergence is consistently within the Fortunian, with the upper (older) 95% highest posterior density (HPD) interval being 541.3 Ma. This inferred origin of trilobites sometime after the Ediacaran–Cambrian boundary represents a very conservative maximum age for the group. Because this analysis did not deliberately impose any node age constraints, the rates and divergence dates for the basal portion of the tree preceding the oldest species analyzed (519 Ma) are necessarily extrapolated from estimates derived from the subsequent (preserved) trilobite fossil record. If evolutionary rates were faster before the first trilobites appear as body fossils, then the inferred rates before 519 Ma in our trees will be underestimates, and the inferred dates would be overestimates. Faster evolutionary rates would allow for the initial phenotypic disparity of trilobite fossils to be established in less time. Imposing a root node age constraint of 522 Ma (13) predictably increased rates at the base of the tree preceding the oldest fossils analyzed (519 Ma), but rates in the early, middle, and late Cambrian remained very similar (*SI Appendix*, Table S1).

Duration of the Cambrian Explosion

Unexpectedly homogeneous rates of morphological evolution throughout the entire Cambrian trilobite fossil record support the idea that the explosion represents a truly brief evolutionary burst that began in the Terreneuvian (at the latest) and had largely concluded by Series 2 (6, 8–10, 24). Regardless of the potential biological and analytical factors responsible for fast initial rates (1, 7, 8, 25), our results provide compelling quantitative evidence that this burst had ended by 519 Ma. This time constraint is also exemplified by the well-established diversity of eumetazoans in the Chengjiang biota of China (26), which has a maximum age of 518.03 ± 0.69/0.71 Ma (27). In fact, from Cambrian Series 2 onward, taxonomic conservatism is apparent among shelly and soft-bodied faunas, further suggesting unremarkable evolutionary rates during this interval; for example, the many shared families and genera across Series 2 and Miaolingian Konservat-Lagerstätten (1, 26). Although the homogeneous rates across the Cambrian are here interpreted to indicate a rapid attainment of postexplosion normality, there is an alternative interpretation: that rates were elevated across most of the Cambrian. However, the longevity of trilobite morphotypes (e.g., genera and families) across the Cambrian (28) and general stability of faunas discussed above make the alternative interpretation less likely. Thus, Chengjiang and younger Cambrian Burgess Shale-type (BST) deposits should not be considered snapshots of the unfolding explosion but rather the early (post-explosion) records of modern-style marine ecosystems.

Despite ongoing debate over the true origins of animal phyla, our data, as well as the Ediacaran–Cambrian geochemical, body, and trace fossil records (1, 3, 9), indicate that a modern-style marine biosphere was fully established by Series 2, followed by broad-scale evolutionary stasis throughout the remainder of the Cambrian. Given the apparent paucity of unequivocal eumetazoan representatives in the Ediacaran (4, 29), it seems that many stem- and crown-group members of most bilaterian phyla had definitively appeared and diversified in ∼20 My (Fig. 4) (8, 9, 29, 30). Among these novel body plans is rampant convergence in various forms of biomineralization (24, 30) and other anatomical innovations that allowed animals increased mobility and ways of sensing their environment (1, 8). Notwithstanding the patchy Terreneuvian fossil record, it is clear that a new style of ecological network—including greatly expanded food webs and associated nutrient cycling, plus complex tiering above and below the substrate that helped reengineer the marine ecosystem—rapidly emerged during this interval (1).

A Cambrian Origin for Trilobites

A Terreneuvian origin for trilobites contradicts previous inferences of a protracted Precambrian history (2, 14–16). Furthermore, our results are supported by evidence from the euarthropod trace fossil record and molecular clocks (Fig. 4). The oldest trilobite-like traces (e.g., *Rusophycus*) are early Fortunian in age (5, 9), and recent molecular clocks (e.g., ref. 31) place the origin of euarthropods in the late Ediacaran or earliest Cambrian. A conservative late Ediacaran root age for euarthropods still permits a

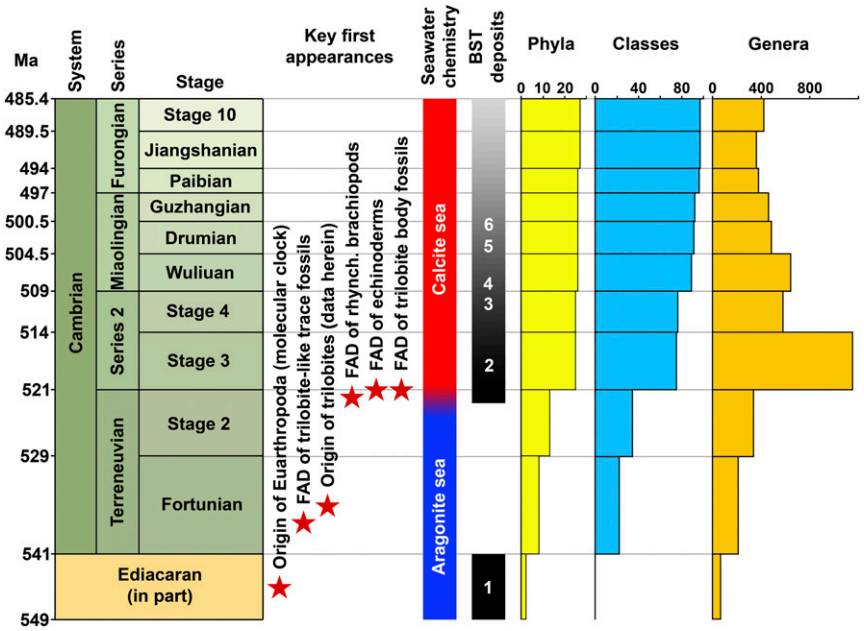


Fig. 4. Key records of early animal evolution, seawater chemistry, and exceptional fossil preservation during the latest Ediacaran to Cambrian. Temporal range of BST deposits (5, 19) shows important examples in ascending stratigraphic order: 1, Khatyspyt Formation (Siberia); 2, Chengjiang (China) and Sirius Passet (North Greenland); 3, Guanshan (China); 4, Burgess Shale (Canada) and Kaili Formation (China); 5, Wheeler Formation (United States); 6, Marjum Formation (United States). BST deposits are less common by the Furongian. Other data sources are as follows: molecular clock date for origin of Euarthropoda represents the posterior mean node age (figure 6 in ref. 31); date for origin of trilobites is the mean node age of the epoch clock model (Fig. 1); first appearance datum (FAD) for trilobite-like traces (5), rhynchonelliform brachiopods (24), echinoderms (33), and trilobite body fossils (13); seawater chemistry (30, 34); and diversity of phyla (3), classes (3), and genera (10).

Cambrian origin for trilobites, given their derived phylogenetic position within Euarthropoda (32).

The absence of Terreneuvian trilobite body fossils can be explained under two potential scenarios. The first scenario is that the fossil record is a reasonably accurate representation of early trilobite evolution, implying that rates of morphological evolution before 519 Ma were substantially faster than subsequently (*SI Appendix*, Table S1). This hypothesis forces rapid dispersal between widely separated paleocontinents and is difficult to reconcile with the provincialism observed in the earliest trilobites. However, if correct, this may also explain the perceived diachronism of the first trilobite fossils on different paleocontinents (12, 13), with the group potentially originating and rapidly radiating out from Siberia, West Laurentia, or West Gondwana (15, 16). The second scenario—more consistent with our results, plus trace fossil, molecular clock, and biogeographic data (5, 11)—is that the earliest trilobites (pre-521 Ma) have not been preserved or yet discovered in Terreneuvian rocks. The diversity of other skeletonized animals from a range of environments throughout the Terreneuvian (24, 30) indicates an adequate shelly fossil record. Thus, the absence of trilobites and indeed other euarthropod body fossils in Terreneuvian rocks could be explained by their nonbiomineralized exoskeletons and the unusual dearth of soft-tissue preservation (especially BST deposits) for this time interval (Fig. 4) (4, 5, 19).

The existence of nonbiomineralized trilobites in the Terreneuvian would have required multiple lineages to simultaneously converge upon a calcite exoskeleton at around 521 Ma, unless initial evolutionary rates were much faster, thus bringing their origin and fewer lineages closer to the lower boundary of Series 2 (in support of the first scenario discussed above). Synchronous biomineralization across two or more trilobite lineages is consistent with the observation that other disparate bilaterians, such as echinoderms and rhynchonelliform brachiopods, also acquired calcitic skeletons around this time (Fig. 4) (24, 33). Notably, this time coincides with a change in ocean chemistry, particularly the onset of a calcite sea (30, 34); ambient seawater chemistry influences the type of biomineral secreted at the time skeletons evolved de novo in a clade (34). These repeated patterns suggest that compelling environmental [e.g., changing Mg/Ca ratios and oxygen levels (1, 30, 35)] and/or biological factors [e.g., predation (30, 36)] were influencing this major episode of biomineralization and diversification during the final stages of the Cambrian explosion. The shelly fossil record of animals thus dramatically improves only around 521 Ma, but by that stage, the Cambrian explosion was largely over.

ACKNOWLEDGMENTS. We thank N. Campione, R. Gaines, L. Holmer, R. Lerosey-Aubril, G. Mángano, and S. Zamora for discussions and feedback; S. Gon III for trilobite drawings; and G. Budd and N. Hughes for constructive reviews. J.R.P. was supported by an Australian Research Council Future Fellowship (FT120100770).

1. Erwin DH, Valentine JW (2013) *The Cambrian Explosion–The Construction of Animal Biodiversity* (Roberts and Company, Greenwood Village, CO).
2. Darwin C (1859) *On the Origin of Species by Means of Natural Selection* (John Murray, London).
3. Erwin DH, et al. (2011) The Cambrian conundrum: Early divergence and later ecological success in the early history of animals. *Science* 334:1091–1097.
4. Cunningham JA, Liu AG, Bengtson S, Donoghue PCJ (2017) The origin of animals: Can molecular clocks and the fossil record be reconciled? *BioEssays* 39:1–12.
5. Daley AC, Antcliffe JB, Drage HB, Pates S (2018) Early fossil record of Euarthropoda and the Cambrian explosion. *Proc Natl Acad Sci USA* 115:5323–5331.
6. Bowring SA, et al. (1993) Calibrating rates of early Cambrian evolution. *Science* 261: 1293–1298.
7. Lee MSY, Soubrier J, Edgecombe GD (2013) Rates of phenotypic and genomic evolution during the Cambrian explosion. *Curr Biol* 23:1889–1895.
8. Budd GE, Jackson ISC (2016) Ecological innovations in the Cambrian and the origins of the crown group phyla. *Philos Trans R Soc Lond B Biol Sci* 371:20150287.
9. Mángano MG, Buatois LA (2014) Decoupling of body-plan diversification and ecological structuring during the Ediacaran-Cambrian transition: Evolutionary and geobiological feedbacks. *Proc Biol Sci* 281:20140038.
10. Na L, Kiessling W (2015) Diversity partitioning during the Cambrian radiation. *Proc Natl Acad Sci USA* 112:4702–4706.
11. Álvaro JJ, et al. (2013) Global Cambrian trilobite palaeobiogeography assessed using parsimony analysis of endemicity. *Early Palaeozoic Biogeography and Palaeogeography*, Geological Society, London, Memoirs, eds Harper DAT, Servais T (Geol Soc, London), Vol 38, pp 273–296.

12. Landing E, Geyer G, Brasier MD, Bowring SA (2013) Cambrian evolutionary radiation: Context, correlation, and chronostratigraphy—Overcoming deficiencies of the first appearance datum (FAD) concept. *Earth Sci Rev* 123:133–172.
13. Zhang X-L, et al. (2017) Challenges in defining the base of Cambrian Series 2 and Stage 3. *Earth Sci Rev* 172:124–139.
14. Fortey RA, Briggs DEG, Wills MA (1996) The Cambrian evolutionary 'explosion': Decoupling cladogenesis from morphological disparity. *Biol J Linn Soc Lond* 57:13–33.
15. Lieberman BS (2002) Phylogenetic analysis of some basal early Cambrian trilobites, the biogeographic origins of the Eutrilobita, and the timing of the Cambrian radiation. *J Paleontol* 76:692–708.
16. Meert JG, Lieberman BS (2004) A palaeomagnetic and palaeobiogeographical perspective on latest Neoproterozoic and early Cambrian tectonic events. *J Geol Soc London* 161:477–487.
17. Gavryushkina A, et al. (2017) Bayesian total-evidence dating reveals the recent crown radiation of penguins. *Syst Biol* 66:57–73.
18. Adrain JM (2011) Class Trilobita Walch, 1771. *Animal Biodiversity: An Outline of Higher-Level Classification and Survey of Taxonomic Richness*, Zootaxa, ed Zhang Z-Q (Magnolia Press, Auckland, New Zealand), Vol 3148, pp 104–109.
19. Peng S, Babcock LE, Cooper RA (2012) The Cambrian Period. *The Geologic Timescale 2012*, eds Gradstein FM, Ogg JG, Schmitz M, Ogg G (Elsevier B.V., Amsterdam), pp 437–488.
20. Barido-Sottani J, et al. (2018) Taming the BEAST—A community teaching material resource for BEAST 2. *Syst Biol* 67:170–174.
21. Suchard MA, et al. (2018) Bayesian phylogenetic and phylodynamic data integration using BEAST 1.10. *Virus Evol* 4:vey016.
22. Lieberman BS (2001) A test of whether rates of speciation were unusually high during the Cambrian radiation. *Proc Biol Sci* 268:1707–1714.
23. Webster M (2007) A Cambrian peak in morphological variation within trilobite species. *Science* 317:499–502.
24. Kouchinsky A, et al. (2012) Chronology of early Cambrian biomineralization. *Geol Mag* 149:221–251.
25. Budd GE, Mann RP (2018) History is written by the victors: The effect of the push of the past on the fossil record. *Evolution* 72:2276–2291.
26. Hou X-G, et al. (2017) *The Cambrian Fossils of Chengjiang, China* (Wiley Blackwell, Oxford), 2nd Ed.
27. Yang C, Li X, Zhu M, Condon DJ, Chen J (2018) Geochronological constraint on the Cambrian Chengjiang biota, South China. *J Geol Soc London* 175:659–666.
28. Whittington HB, et al. (1997) *Treatise on Invertebrate Paleontology, Part O, Revised. Arthropoda 1, Trilobita 1 (Introduction, Order Agnostida, Order Redlichiida)* (Geol Soc Am, Boulder, CO).
29. Budd GE, Jensen S (2000) A critical reappraisal of the fossil record of the bilaterian phyla. *Biol Rev Camb Philos Soc* 75:253–295.
30. Wood R, Zhuravlev AY (2012) Escalation and ecological selectively of mineralogy in the Cambrian radiation of skeletons. *Earth Sci Rev* 115:249–261.
31. dos Reis M, et al. (2015) Uncertainty in the timing of origin of animals and the limits of precision in molecular timescales. *Curr Biol* 25:2939–2950.
32. Legg DA, Sutton MD, Edgecombe GD (2013) Arthropod fossil data increase congruence of morphological and molecular phylogenies. *Nat Commun* 4:2485.
33. Smith AB, Zamora S, Álvaro JJ (2013) The oldest echinoderm faunas from Gondwana show that echinoderm body plan diversification was rapid. *Nat Commun* 4: 1385.
34. Porter SM (2010) Calcite and aragonite seas and the *de novo* acquisition of carbonate skeletons. *Geobiology* 8:256–277.
35. Peters SE, Gaines RR (2012) Formation of the 'Great Unconformity' as a trigger for the Cambrian explosion. *Nature* 484:363–366.
36. Bicknell RDC, Paterson JR (2018) Reappraising the early evidence of durophagy and drilling predation in the fossil record: Implications for escalation and the Cambrian explosion. *Biol Rev Camb Philos Soc* 93:754–784.

PNAS
www.pnas.org

Supporting Information (SI)

Trilobite evolutionary rates constrain the duration of the Cambrian explosion

John R. Paterson, Gregory D. Edgecombe, Michael S.Y. Lee

Contents

- Materials and Methods:
    1. Morphological data
        a. Taxonomic and character sampling
        b. Discrete characters
        c. Meristic characters
        d. Continuous characters
    2. Phylogenetic analyses
        a. Bayesian tip-dated analyses
        b. Parsimony analysis
        c. Implications for trilobite systematics
- References
- Table S1
- Figs S1 to S3

Other Supporting Information includes the following 5 data files on Dryad Digital Repository: https://doi.org/10.5061/dryad.v7q827k

- Dryad Dataset 1: Taxon list, associated stratigraphic/age data, and references for character codings (Excel format)
- Dryad Dataset 2: Discrete character matrix (Nexus format)
- Dryad Dataset 3: Meristic and continuous character matrices (Excel format)
- Dryad Dataset 4: BEAST and BEAST2 executable files (including formatted data) in xml format, and output trees in Nexus/FigTree format. Multiple files (zipped)
- Dryad Dataset 5: TNT executable files (including formatted data) in plain text format, and output tree in Nexus/FigTree format. Multiple files (zipped)

www.pnas.org/cgi/doi/10.1073/pnas.1819366116

Materials and Methods

1. Morphological data

1a. Taxonomic and character sampling

The new phylogenetic dataset is the largest and most comprehensive for trilobites compiled to date (23, 37), comprising 107 Cambrian species (ranging from Series 2 to the Furongian [519–486 Mya, for sampled taxa]) and 115 exoskeletal characters (107 discrete, 2 meristic, 6 continuous) (Dryad Datasets 1 to 3). Ideally, tip-dated Bayesian analyses should utilize datasets where all variable characters, including autapomorphies (unique changes on terminal branches), are sampled with equal intensity (38, 39). The current dataset was explicitly compiled for tip-dated analyses, unlike the majority of datasets collected under a cladistics framework, which under-sample or totally exclude autapomorphies because they are parsimony-uninformative. The discrete characters used here represent all variable characters identified, with autapomophic variation sampled with the same thoroughness as synapomorphic variation. The meristic and continuous datasets, by their nature, also automatically encapsulate changes in internal (synapomorphic) as well as terminal (autapomorphic) branches: the measurements obtained for terminal taxa are the result of both types of change.

In order to cover all major Cambrian trilobite morphotypes and diversity, at least one exemplar species from most valid Cambrian families was chosen (Dryad Dataset 1), following the latest classification schemes (18, 40). We follow the most recent suprafamilial classification of Trilobita (18) in excluding the suborder Agnostina (*sensu* [41]) based on morphological evidence—particularly regarding the non-lamellipedian appendages of *Agnostus pisiformis* (42), the atypical ‘hypostome’ of various agnostoids (43), and the general disparity of dorsal exoskeletal features between agnostoids and other trilobites, including eodiscoids (44)—which suggests that this group is more closely allied to crustaceans than to artiopodans (see 32, 45-50). Where possible, representative taxa were preferentially selected based on the availability of well-preserved articulated exoskeletons or silicified sclerites, as well as known ontogenies, to maximize characters and minimize missing data. Sources of character codings are presented in Dryad Dataset 1.

Associated stratigraphic data (including biozones) for each species were compiled from the primary literature. Numerical (absolute) ages (Ma) used in the Bayesian tip-dated analyses (discussed below) were obtained by correlating regional biozones and stages with the calibrated 2012 Cambrian timescale (19) and subsequent updates to the International Chronostratigraphic Chart (http://www.stratigraphy.org/index.php/ics-chart-timescale). In some cases, correlations were supplemented using charts in key biostratigraphic references (e.g. 12, 13, 51-56). The dates are also presented in Dryad Dataset 1.

The character set (presented below and in Dryad Datasets 2 and 3) is a modified and greatly expanded version of the matrix developed by Paterson & Edgecombe (57). It includes discrete characters that have commonly been used in higher-level classification and are often considered to diagnose clades (e.g., families, superfamilies, suborders and orders) (37, 41, 58), for example: absence or presence of dorsal facial sutures (59, 60); modes of hypostomal attachment (61, 62); development of the ventral median cephalic suture (63-68); fulcrate or nonfulcrate thoracic articulation (69); forms of trunk tagmosis, particularly the division into pro- and opisthothorax (57, 70); and a variety of ontogenetic characters, such as trends in thoracic macropleurality of meraspides and holaspides (71). As noted above, many of the discrete

character states are autapomorphies, which are required for tip-dating methods to accurately estimate amounts and rates of morphologic change across every branch and, in turn, divergence dates.

Meristic and continuous characters are also included directly (rather than recoded into a small number of discrete states) to provide a more objective approach to analyzing highly variable traits that are also considered important in trilobite systematics, such as the number of thoracic segments, or the relative size of the pygidium compared to the cephalon (58). Categorization of the latter trait, for example, is often simplified to the somewhat subjective terms 'micropygous' (the pygidium is shorter (sag., exsag.) and narrower (tr.) than the cephalon), 'isopygous' (a pygidium of about the same width (tr.) and length (sag., exsag.) as the cephalon), and 'macropygous' (the maximum pygidial width (tr.) exceeds that of the cephalon) (72). Treating such a trait as continuous using actual measurements thus avoids the use of arbitrary subdivisions to formulate discrete states, such as gap coding.

1b. Discrete characters

Discrete characters 1–90 are morphological features of the holaspid (adult) exoskeleton organized into an anterior-posterior arrangement (1–68, cephalic; 69–79, thoracic; 80–90, pygidial), and characters 91–107 are ontogenetic. Codings for all species are available in Dryad Dataset 2. Where relevant, all autapomorphies are noted under each character; it is important to state that these autapomorphies have the potential to become synapomorphies by including additional taxa or revising unknown ('?') character state codings with new material of species in the current dataset.

1. Dorsal ecdysial cephalic sutures:

[See Paterson & Edgecombe's (57) character 1 for discussion]

0. absent
1. present

2. Dorsal ecdysial suture pattern:

[Taxa coded as state 0 for character 1 are coded as inapplicable for this character. State 2 represents an autapomorphy of *Yunnanocephalus yunnanensis*]

0. opisthoparian
1. proparian
2. gonatoparian

3. Opisthoparian suture pattern:

[Taxa coded as states 1 or 2 (or inapplicable) for character 2 are coded as inapplicable for this character. The following states represent autapomorphies: 9 (*Akoldinioidia shanjiangensis*); 11 (*Asioptychaspis subglobosa*); 13 (*Lisania paibiensis*); 14 (*Conocoryphe sulzeri*); 15 (*Tasmacephalus platypus*)]

0. gamma to alpha points long, straight or slightly curved, and divergent anteriorly; epsilon to omega points long and strongly divergent (defining narrow (exsag.) posterolateral projections)
1. gamma to alpha points very short, straight, and subparallel or slightly convergent; epsilon to omega points very short

2. gamma to beta points divergent anteriorly or subparallel, but with slight adaxial bend at beta points, with very short section between converging beta and alpha points; epsilon to omega points very short
3. gamma to beta points long and divergent anteriorly or subparallel, but with distinct adaxial bend at beta points, with longer section between converging beta and alpha points; epsilon to omega points long and strongly divergent (terminating at pointed posterolateral projections)
4. gamma to beta points divergent anteriorly, but with strong adaxial bend at beta points, with short section between converging beta and alpha points; epsilon to omega points long and strongly divergent (terminating at pointed posterolateral projections)
5. gamma to beta points divergent anteriorly, but with strong adaxial bend at beta points, with short section between converging beta and alpha points; epsilon to omega points long and divergent at about 180 degs. before curving posteriorly (terminating at rounded posterolateral projections)
6. gamma to alpha points straight, and subparallel or convergent anteriorly; epsilon to omega points very long and moderately to strongly divergent
7. gamma to alpha points curved and convergent anteriorly; epsilon to omega points moderately long and moderately divergent
8. gamma to alpha points broadly curved and convergent anteriorly; epsilon to omega points long and widely divergent before curving posteriorly (terminating at broadly rounded posterolateral projections)
9. alpha to omega submarginal and gently sinuous
10. gamma to beta points divergent anteriorly with adaxial bend at beta points, with very short section between converging beta and alpha points; epsilon to omega points long and strongly divergent (forming wedge-shaped posterolateral projections)
11. gamma to alpha points very short, straight and subparallel; epsilon to omega points very long, straight and slightly divergent (forming large triangular posterolateral projections)
12. gamma to beta points long and parallel, but with broad adaxial curve from beta to alpha points; epsilon to omega short, curved and moderately divergent
13. gamma to beta points short and subparallel, but with strong adaxial bend at beta points, with short section between converging beta and alpha points; epsilon to omega points very short
14. alpha to omega mostly marginal, but curves over genal angle
15. gamma to alpha points long and sinuous (extending onto frontal 'snout'), epsilon to omega points moderately divergent

4. Development of axial furrows on cephalon:

0. clearly impressed
1. nearly effaced

5. Preoccipital glabellar shape in holaspides:

[This character is modified from Paterson & Edgecombe's (57) character 2]

0. tapers anteriorly with narrow frontal lobe

1. subparallel/subquadrate, with gently rounded or truncated frontal lobe
2. subparallel, with strongly rounded frontal lobe
3. constricted at L1, L2 or L3 with enlarged frontal lobe
4. moderately to strongly expands anteriorly
5. flask-shaped, with inflated L1

6. SO:

[Taxa coded as inapplicable possess a cranidial spine: character 24, state 1. This character is modified from Paterson & Edgecombe's (57) character 5]

0. deepened laterally and shallow medially
1. subequal depth across glabella
2. deep medially, shallowing near axial furrows
3. effaced on testaceous dorsal surface

7. S1:

[This character is modified from Paterson & Edgecombe's (57) character 6]

0. absent or highly effaced in testaceous and exfoliated material
1. confined laterally and connected to axial furrow
2. transglabellar
3. isolated pits connected by medial furrow
4. isolated, abaxially curved furrows

8. Shape of transglabellar S1:

[Taxa coded as states 0, 1, 3 or 4 for character 7 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 7]

0. straight or weakly convex
1. strongly convex posteriorly

9. S1 bifurcates adaxially:

[Taxa coded as states 0, 2, 3 or 4 for character 7 are coded as inapplicable for this character]

0. absent
1. present

10. S2:

[This character is modified from Paterson & Edgecombe's (57) character 8]

0. absent or highly effaced in testaceous and exfoliated material
1. confined laterally
2. transglabellar
3. isolated pits connected by medial furrow

11. Shape of laterally confined S2:

[Taxa coded as states 0, 2 or 3 for character 10 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 9]

0. straight or weakly convex
1. strongly convex anteriorly

2. isolated slots (abaxial portions of L2 and L3 connected)
3. isolated pits connected by medial furrow

12. Orientation of laterally confined S2:
[Taxa coded as states 0, 2 or 3 for character 10 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 10]
0. directed forwards abaxially
1. transverse

13. Shape of transglabellar S2:
[Taxa coded as states 0, 1 or 3 for character 10 are coded as inapplicable for this character]
0. straight or weakly convex
1. strongly convex posteriorly

14. S3:
[This character is modified from Paterson & Edgecombe's (57) character 12]
0. absent or highly effaced in testaceous and exfoliated material
1. confined laterally
2. transglabellar
3. isolated pits connected by medial furrow

15. Shape of laterally confined S3:
[Taxa coded as states 0, 2 or 3 for character 14 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 13]
0. straight or weakly convex
1. strongly convex anteriorly
2. isolated slots

16. Orientation of laterally confined S3:
[Taxa coded as states 0, 2 or 3 for character 14 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 14]
0. directed forwards abaxially
1. directed backwards abaxially
2. transverse

17. Shape of transglabellar S3:
[Taxa coded as states 0, 1 or 3 for character 14 are coded as inapplicable for this character]
0. straight or weakly convex
1. strongly convex posteriorly

18. Strong sagittal glabellar furrow confluent with S1, S2 and S3:
[State 1 represents an autapomorphy of *Sao hirsuta*]
0. absent
1. present

19. Extent of frontal lobe of glabella relative to adjacent position of anteroproximal part of eye (or palpebro-ocular) ridges:

0. frontal lobe does not extend anteriorly beyond eye ridges
1. frontal lobe extends anteriorly beyond eye ridges

20. Medial sulcus in anterior margin of frontal lobe of glabella:

0. absent
1. present

21. Single preoccipital glabellar tubercle:

0. absent
1. present

22. Series of glabellar axial nodes anterior to LO:

[State 1 represents an autapomorphy of *Nephrolenellus geniculatus*]

0. absent
1. present

23. LO with anterior median tubercle and posterior occipital spine:

0. absent
1. present

24. Cranidial spine:

[Definition of the cranidial spine follows Jell ([73], p. 4). This character equates to Cotton and Fortey's (74) character 65]

0. absent
1. present

25. Extended, snout-like frontal margin of cephalon:

[State 1 represents an autapomorphy of *Tasmacephalus platypus*]

0. absent
1. present

26. Length of cephalic border:

[State 1 represents an autapomorphy of *Pterocephalia norfordi*]

0. sagittal length considerably shorter than remainder of cephalon
1. sagittal length equal to or greater than remainder of cephalon

27. Radial scrobicules on cephalic border:

[State 1 represents an autapomorphy of *Pagetia ocellata*]

0. absent
1. present

28. Single row of large, evenly spaced tubercles on anterior and lateral cephalic border:

[State 1 represents an autapomorphy of *Serrodiscus speciosus*]

0. absent
1. present

29. Cephalic border furrow with pits:
[This character relates to the structure described by Adrain (18; footnote 33, p. 107) and illustrated by Whittington (fig. 30.2a, b in 75)]
0. absent
1. present

30. Preglabellar field:
[This equates to Paterson & Edgecombe's (57) character 3]
0. absent
1. present

31. Plectrum on dorsal surface:
[Taxa coded as state 0 for character 30 or state 1 for characters 33 or 35 are coded as inapplicable for this character. Presence of a plectrum (state 1) may also incorporate what is termed a 'median preglabellar ridge' (72). See Paterson & Edgecombe's (57) character 4 for additional discussion]
0. absent
1. present

32. Shape of plectrum:
[Taxa coded as state 0 for characters 30 or 31 (or inapplicable for the latter) are coded as inapplicable for this character]
0. very narrow (tr.), subparallel-sided ridge
1. broad (width >50% that of frontal glabellar lobe), subparallel-sided ridge
2. posteriorly tapered extension of the anterior cephalic border

33. Medial swelling in preglabellar field (= preglabellar boss):
[Taxa coded as state 0 for character 30 or state 1 for characters 31 or 35 are coded as inapplicable for this character]
0. absent
1. present

34. Shape of preglabellar boss:
[Taxa coded as state 0 for characters 30 or 33 (or inapplicable for the latter) are coded as inapplicable for this character. State 0 represents an autapomorphy of *Penarosa netenta* and state 1 represents an autapomorphy of *Strenuaeva inflata*]
0. narrow (tr.), long (sag.) swelling
1. broad (tr.) swelling that merges with anterior border

35. Narrow (slit-like) sagittal furrow on preglabellar field:
[Taxa coded as state 0 for character 30 or state 1 for characters 31 or 33 are coded as inapplicable for this character. State 1 represents an autapomorphy of *Liostracina tangwangzhaiensis*]

0. absent
1. present

36. Anteriorly divergent furrows on preglabellar field emanating from axial furrows:
[Taxa coded as state 0 for character 30 are coded as inapplicable for this character]
0. absent
1. present

37. Extent of anteriorly divergent furrows on preglabellar field:
[Taxa coded as state 0 for characters 30 or 36 (or inapplicable for the latter) are coded as inapplicable for this character]
0. extending to, and connecting with anterior border furrow
1. not extending to anterior border furrow

38. Transversely elongate triangular lobes situated at anterolateral corners of glabella:
[State 1 represents an autapomorphy of *Dinesus ida*]
0. absent
1. present

39. Dorsal eyes:
0. present
1. absent

40. Eye type:
[Taxa coded as state 1 for character 39 are coded as inapplicable for this character]
0. holochroal
1. abathochroal

41. Curvature of palpebral lobe:
[Taxa coded as state 1 for character 39 are coded as inapplicable for this character]
0. gently to moderately curved
1. strongly curved, such that arch from gamma to epsilon is semi-circular

42. Palpebral lobe and eye/ocular ridge:
[Taxa coded as state 1 for character 39 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 17]
0. continuous, undifferentiated
1. distinctly differentiated, based on relative width and/or relief
2. isolated palpebral lobe, eye/ocular ridge absent or highly effaced on testaceous dorsal surface
3. anterior tip of palpebral lobe abuts axial furrow or glabella

43. Confluence of palpebro-ocular ridges (or palpebral lobe) and glabella:
[Taxa coded as state 1 for character 39 or state 2 for character 42 are coded as inapplicable for this character. This character is a modified version of Cotton's (76) character 47]

0. interrupted by axial furrow
1. strongly confluent

44. Parafrontal band:

[Taxa coded as state 1 for character 39 are coded as inapplicable for this character. See Paterson & Edgecombe's (57) character 15 for discussion]

0. absent
1. present

45. Position of posterior tips of palpebral lobe or palpebro-ocular ridge:

[Taxa coded as state 1 for character 39 are coded as inapplicable for this character. State 4 represents an autapomorphy of *Ajrikina hunanensis*. This character is modified from Paterson & Edgecombe's (57) character 18]

0. opposite LO or SO
1. opposite L1 or S1
2. opposite L2 or S2
3. anterior of S2 and posterior of anterior margin of glabella
4. anterior of glabella

46. Fossulae:

[See Paterson & Edgecombe's (57) character 16 for discussion]

0. absent
1. present as small, discrete pits

47. Width (tr.) of interocular area adjacent to posterior tip of palpebro-ocular ridge or palpebral lobe in holaspides:

[Taxa coded as state 1 for character 39 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 19]

0. less than one-third the width (tr.) of the occipital ring
1. one-third to half the width (tr.) of the occipital ring
2. more than half to the entire width (tr.) of the occipital ring
3. more than the entire width (tr.) of the occipital ring

48. Bacculae in holaspides:

0. absent
1. present

49. Size of bacculae:

[Taxa coded as state 0 for character 48 are coded as inapplicable for this character]

0. small, adjacent to SO and/or L1
1. elongate, occupying most of the palpebral area and posterior field of the fixigenae

50. Intergenal ridge:

[This equates to Paterson & Edgecombe's (57) character 23]

0. absent
1. present

51. Intergenal spine:
[This equates to Paterson & Edgecombe's (57) character 21]
0. absent
1. present

52. Depressed fixigenal field, with depression in posterior field widening abaxially:
[State 1 represents an autapomorphy of *Ajrikina hunanensis*]
0. absent
1. present

53. Librigenal field or extraocular area adjacent to the farthest abaxial point of palpebral lobe or palpebro-ocular ridge:
[Taxa coded as state 1 for character 39 are coded as inapplicable for this character]
0. visible in dorsal view
1. not visible in dorsal view due to lobe/ridge overhanging genal field

54. Genal spine:
0. absent
1. present

55. Shape of genal spine:
[Taxa coded as state 0 for character 54 are coded as inapplicable for this character]
0. tapering
1. bulbous

56. Cephalon with strong, coarsely reticulated (honeycomb-like) ornament:
0. absent
1. present

57. Radiating grooves on frontal area:
[Taxa without a frontal area or coded as state 1 for character 56 are coded as inapplicable for this character. State 1 represents an autapomorphy of *Hungaia puelchana*]
0. absent
1. present

58. Cuticular sculpture of irregular star-shaped tubercles:
[State 1 represents an autapomorphy of *Eurekia rintintini*]
0. absent
1. present

59. Prominent fingerprint-like ridges on glabella, fixigenae and librigenae:
[State 1 represents an autapomorphy of *Asioptychaspis subglobosa*]
0. absent
1. present

60. Ventral median cephalic suture in late holaspides:
[Taxa that lack librigenae or a rostral plate are coded as inapplicable. State 2 represents an autapomorphy of *Norwoodia boninoi*]
0. absent (librigenae wholly separated by rostral plate)
1. complete ventral median suture
2. large rostellum anteriorly, ventral median suture posteriorly
3. librigenae fused ventrally (yoked)

61. Mode of development of ventral median cephalic suture:
[Taxa coded as states 0 or 3 or inapplicable for character 60 are coded as inapplicable for this character. This character is based on the modes of development of the ventral median suture proposed by Park & Choi (fig. 9 in 66). State 1 represents an autapomorphy of *Pterocephalia norfordi*]
0. developing from triangular rostellum-like sclerite posteriorly
1. developing from triangular rostellum anteriorly
2. developing late in ontogeny from yoked librigenae

62. Hypostomal attachment:
[See Paterson & Edgecombe's (57) character 24 for discussion]
0. natant
1. conterminant

63. Hypostome fused to cephalic doublure in holaspides:
[Taxa coded as state 0 for character 62 are coded as inapplicable for this character]
0. absent (= functional hypostomal suture)
1. present (= hypostome fused to rostral plate)

64. Width (tr.) of hypostomal lateral border:
0. narrower than middle body
1. wider than middle body

65. Maculae:
0. absent (or poorly developed)
1. present

66. Form and position of maculae:
[Taxa coded as state 0 for character 65 are coded as inapplicable for this character. State 2 represents an autapomorphy of *Zacanthopsis palmeri*]
0. elevated, narrow ridges immediately posterior to deep, lateral middle furrows
1. elevated, subovate platforms immediately posterior to lateral middle furrows or pits
2. elevated, subovate platforms anterior of middle furrow

67. Isolated U-shaped middle furrow on middle body of hypostome:
0. absent
1. present

68. Posterior lobe of middle body of hypostome with posteriorly-directed protuberance:
0. absent
1. present

69. Tagmosis of thorax:
[This character is modified from Paterson & Edgecombe's (57) character 25]
0. pro- and opisthothorax undifferentiated
1. pro- and opisthothorax differentiated

70. Prothoracic/opisthothoracic boundary marked by macropleural segment:
[Taxa coded as state 0 for character 69 are coded as inapplicable for this character]
0. absent (or decoupled when macropleural segment present)
1. present

71. Macropleural thoracic spine(s):
0. absent
1. present

72. Position of macropleural thoracic spine(s) developed on:
[Taxa coded as state 0 for character 71 are coded as inapplicable for this character. The following states represent autapomorphies: 0 (*Hamatolenus vincenti*); 3 (*Burnetiella leechi*); 5 (*Palaeadotes hunanensis*). This character is modified from Paterson & Edgecombe's (57) character 32]
0. second segment
1. third segment
2. fourth segment
3. fifth segment
4. sixth segment
5. seventh segment
6. eleventh segment

73. Inner portion of pleura on segment bearing macropleural thoracic spine(s):
[Taxa coded as state 0 for character 71 are coded as inapplicable for this character]
0. does not strongly expand abaxially [= 'amplipleural' *sensu* Palmer (77)]
1. strongly expands abaxially, distorting adjacent posterior segment(s) [= 'macropleural' *sensu* Palmer (77)]
2. strongly expands abaxially, distorting adjacent anterior and posterior segments [= 'hyperpleural' *sensu* Palmer (77)]

74. Two batches of thoracic segments defined by differences in pleural tips separated by macropleural spines:
[State 1 represents an autapomorphy of *Burnetiella leechi*]
0. absent
1. present (anterior batch with falcate pleural spines, posterior batch with spatulate tips)

75. Thoracic segments 5 and 6 fused:
[Taxa with less than six thoracic segments (i.e., eodiscoids and *Akoldinioidia shanjiangensis*) are coded as inapplicable. See Paterson & Edgecombe's (57) character 26 for discussion]
0. free
1. fused

76. Hypertrophied thoracic axial spine(s), relative to other spines:
[The following states represent autapomorphies: 1 (*Norwoodia boninoi*); 2+5 (*Redlichia takooensis*); 3 (*Bolaspidella housensis*); 4 (*Eoredlichia intermediata*); 6 (*Montezumaspis parallela*); 8 (*Elliptocephala asaphoides*)]
0. absent
1. present on T4
2. present on T6
3. present on T8
4. present on T9
5. present on T11
6. present on T13
7. present on T15
8. present on T16 to T18

77. Fulcrum on thoracic pleurae:
[Taxa with unknown thoracic morphology are coded as being fulcrate (state 1) if there is an articulating facet on the pygidium. See Paterson & Edgecombe's (57) character 27 for discussion]
0. absent
1. present

78. Thoracic pleural tips:
[See Paterson & Edgecombe's (57) character 29 for discussion]
0. falcate/sentate spines (pleural margins continuous onto spines)
1. "thorn-like" spines (base of spine much narrower (exsag.) than pleural region)
2. spatulate (non-spinose)

79. Length of pleural spines along thorax (excluding macropleural spines):
[Taxa coded as state 2 for character 78 are coded as inapplicable for this character. See Paterson & Edgecombe's (57) character 30 for discussion]
0. of equal length or slightly increasing or decreasing posteriorly
1. strongly increasing posteriorly

80. Differentiation of pygidial pleurae in late holaspides:
[State 1 includes taxa that have a distinct pygidial border]
0. anteriormost pleural segment resembles unreleased thoracic segment in similar orientation and expression of pleural and interpleural furrows and distal tips, such that the thoracic-pygidial boundary is indistinct

1. anteriormost pleura (or pleural region) differs from thoracic pleural morphology in the orientation and/or expression of pleural and interpleural furrows and distal tips
2. anteriormost pleural segment is macropleural, differing significantly from thoracic segments and more posterior pygidial segments

81. Pygidial pleural furrows:

0. well impressed from axial furrow to pygidial margin
1. well impressed near axial furrow, but become shallower or effaced near pygidial margin, including termination at border furrow
2. weakly developed or effaced

82. Bilobate pygidial terminal piece:

[See Paterson & Edgecombe's (57) character 35 for discussion]

0. absent
1. present

83. Postaxial ridge on pygidium:

0. absent
1. present

84. Extent of postaxial ridge on pygidium:

[Taxa coded as state 0 for character 83 are coded as inapplicable for this character]

0. extends to margin
1. merges with border or terminal area (before margin)

85. Vertical, wall-like distal pygidial pleural area:

[State 1 represents an autapomorphy of *Notchpeakia taylori*]

0. absent
1. present

86. Posteromedial margin of pygidium immediately behind axis (in dorsal view):

[State 4 represents an autapomorphy of *Lonchopygella megaspina*. See Paterson & Edgecombe's (57) character 36 for discussion]

0. smooth (non-spinose), continuously rounded
1. smooth (non-spinose) with shallow, rounded concavity
2. with deep, acutely v-shaped notch
3. concave and bounded by pair of posterolateral marginal spines
4. extends into single, large axial spine

87. Paired marginal pygidial spines in late holaspides:

[This equates to Paterson & Edgecombe's (57) character 37]

0. absent
1. present

88. Marginal pygidial spine morphology in late holaspides:

[Taxa coded as state 0 for character 87 are coded as inapplicable for this character. The following states represent autapomorphies: 5 (*Tricrepicephalus texanus*); 6 (*Yinites wanshougongensis*); 7 (*Dikelocephalus minnesotensis*); 8 (*Zhangshania typica*). This character is modified from Paterson & Edgecombe's (57) character 38]

0. single pair of short, anterolateral spines
1. single pair of short, posterolateral spines
2. two or more pairs of similar spines only
3. single pair of long spines extending from first macropleural segment, and remainder of pygidial margin smooth
4. single pair of long spines extending from first macropleural segment, and remainder of pygidial margin with short spines
5. single pair of long spines extending from posterior portion of pleural field
6. two short anterolateral spines, plus a single pair of long spines that are decoupled from pleural segmentation and extend from posterolateral margin of pygidium
7. single pair of short, posterolateral spines that are decoupled from pleural segmentation
8. single pair of long spines that are decoupled from pleural segmentation, with outer edge of spine extending from anterolateral margin of pygidium

89. Macropleural pygidial spine:

[Taxa coded as state 0 for character 87 are coded as inapplicable for this character]

0. absent
1. present

90. Position of macropleural pygidial spine in late holaspides:

[Taxa coded as state 0 for characters 87 or 89 are coded as inapplicable for this character. State 1 represents an autapomorphy of *Tricrepicephalus texanus*]

0. anteriormost segment
1. posterior portion of pleural field

91. Calcified protaspid stage:

[See Paterson & Edgecombe's (57) character 39 for discussion]

0. absent
1. present

92. Anterior cranidial/cephalic border in protaspides and/or early meraspides:

[See Paterson & Edgecombe's (57) character 41 for discussion]

0. absent
1. present

93. Procranidial spines in early meraspides:

[State 1 represents an autapomorphy of *Bristolia anteros*. See Paterson & Edgecombe's (57) character 42 for discussion]

0. absent
1. present

94. Form of frontal lobe of glabella in protaspides and/or early meraspides:
[This character is modified from Paterson & Edgecombe's (57) character 43]
0. bulbous and distinctly wider (tr.) than posterior portion of glabella
1. lateral margins contiguous with remainder of glabella, typically with slight taper or expansion anteriorly
2. strongly expanding anteriorly, as defined by diverging axial furrows

95. Longitudinal medial glabellar furrow in protaspides:
[Taxa coded as state 0 for character 91 are coded as inapplicable for this character]
0. absent
1. present

96. Protaspides with prominent protoaxial node/ring:
[Taxa coded as state 0 for character 91 are coded as inapplicable for this character. Definition of the protoaxial node follows Cederström et al. (78)]
0. absent
1. present

97. Transverse furrows on interocular cheeks in protaspides and/or early meraspides:
[See Paterson & Edgecombe's (57) character 40 for discussion]
0. absent
1. present

98. Protaspides with protobacculae:
[Taxa coded as state 0 for character 91 are coded as inapplicable for this character. Definition of protobacculae follows Cederström et al. (78)]
0. absent
1. present

99. Intergenal/fixigenal spines in protaspides and/or early meraspides:
[This equates to Paterson & Edgecombe's (57) character 44]
0. absent
1. present

100. Protogenal lobes in protaspides:
[Taxa coded as state 0 for character 91 are coded as inapplicable for this character. Definition of protogenal lobes follows Cederström et al. (78)]
0. absent
1. present

101. Protaspides with pair of prominent lobes on posterior margin:
[Taxa coded as state 0 for character 91 are coded as inapplicable for this character]
0. absent
1. present

102. Hypostome and rostral plate fused in early meraspides:

[Eodiscoids are coded as inapplicable, as they lack a rostral plate. State 1 represents an autapomorphy of *Zacanthopsis palmeri*]

0. absent
1. present

103. Immature hypostome with inflated anterior lobe of middle body and spinose "frill" with six spine pairs on posterior margin:

0. absent
1. present

104. Thoracic macropleurality in meraspides:

0. absent
1. present

105. Development of thoracic macropleurality from meraspides to holaspides:

[Taxa coded as state 0 for character 104 are coded as inapplicable for this character. This character is modified from Paterson & Edgecombe's (57) character 45]

0. macropleural segment/s retained in late holaspides
1. macropleural segment/s lost in late holaspides

106. Large pair of anterolateral transitory pygidial or pygidial spines in late meraspides and early holaspides, respectively, but lost in late holaspides:

0. absent
1. present

107. Trunk segmentation schedule:

[*Neocobboldia chinlinica* has a partial uncertainty coding (state 1 or 2), following Hughes et al. (79)]

0. protarthrous (holaspid phase precedes epimorphic phase)
1. synarthromeric (synchronous holaspid and epimorphic phases)
2. protomeric (epimorphic phase precedes holaspid phase)

1c. Meristic characters

Quantitative (raw) data for the meristic characters can be found in Dryad Dataset 3.

108 (M1). Maximum number of thoracic segments in late holaspides

109 (M2). Maximum number of axial rings in pygidium in late holaspides

[This excludes the articulating half ring and terminal piece and is based on the largest known pygidium for each species. For some species, the pygidial axis is largely effaced, but segments can sometimes be discerned, particularly on the ventral surface or exfoliated specimens]

1d. Continuous characters

Quantitative (raw) data for the continuous characters can be found in Dryad Dataset 3. Details about measurements and calculations are discussed below and can also be found in Fig. S1.

110 (C1). Maximum sagittal exoskeletal length (mm)
[Where possible, maximum length is measured from the anterior (sag.) margin of the cephalon to the posterior (sag.) margin of the pygidium (excluding axial spines that may extend beyond the pygidium in some species). In some cases, the maximum length has been measured on the largest known articulated specimen, but then extrapolated using the largest known cephalon/cranidium or pygidium]

111 (C2). Cephalic/cranidial:pygidial area ratio
[This ratio was determined by dividing the cephalic/cranidial area by the pygidial area – see Fig. S1 for further details. Where possible, measurements were recorded from the largest known and/or most complete, articulated exoskeletons, but are not necessarily associated with the exoskeletal length recorded for character 110. Otherwise, each measurement was taken from the largest known cephalon/cranidium and pygidium of each taxon, provided there is a sensible match in axial width between the occipital ring and first pygidial axial ring (i.e., the latter being roughly equal or narrower). This ratio avoids use of the subjective terms 'micropygous', 'isopygous' and 'macropygous']

112 (C3). Ratio of thoracic:pygidial segments in trunk
[This was calculated using data from meristic characters 108 and 109]

113 (C4). Length of palpebral lobe from γ to ε (as % of sag. cephalic/cranidial length)
[Recorded percentage represents the mean value based on holaspid specimens. Cephalic/cranidial length excludes the occipital spine. Blind taxa are coded as inapplicable]

114 (C5). Width (tr.) of thoracic axial ring on T1 (as % of the entire width of T1)
[Recorded percentage represents the mean value based on holaspid specimens]

115 (C6). Maximum width (tr.) of pygidial axis (as % of max. pygidial width)
[Recorded percentage represents the mean value based on holaspid specimens]

2. Phylogenetic analyses

The discrete, meristic and continuous character data were analyzed simultaneously using tip-dated Bayesian approaches, as well as maximum parsimony.

2a. Bayesian tip-dated analyses

The Bayesian tip-dated analyses (17, 80, 81) were performed using BEAST (21), with BEAST2 (82) also used when appropriate. The data were explicitly gathered to satisfy the

assumptions of tip dating, namely in sampling all variable traits, including autapomorphies, with equal intensity (see above). The optimal evolutionary histories that explain the discrete, meristic and continuous traits, as well as stratigraphic dates, were inferred using Markov-chain Monte Carlo (MCMC) approaches, as implemented in BEAST 1.8.4 with post-processing modules LogCombiner and TreeAnnotator (83) and OutbreakTools (84). In the context of this analysis, BEAST was preferred over BEAST2 because it can implement epoch-clock models (85). In contrast, BEAST2 could not implement any analytical models over those available in BEAST; the SABD tree prior could not be implemented on this dataset (see below). The executable XML files, with annotations describing model, prior, MCMC and logging settings, are available in Dryad Dataset 4. The description below summarizes the major points, focusing on the character data and clock models.

Tip calibrations were employed using a point stratigraphic range for each taxon. In all main analyses, there was no constraint on minimum or maximum root age; this was estimated entirely by the data and model. Thus, the age of the MRCA of all (sampled) trilobites was entirely extrapolated based on the age and disparity of the included fossils, and their inferred rates of phenotypic divergence. As discussed in the main text, this very likely provides a *maximum* estimated age for trilobites. Additional analyses were also undertaken with the root age constrained to be 522 Ma (see below). Trees were rooted with Fallotaspididae based on their basal position in previous phylogenetic analyses (e.g., 57, 86, 87).

The discrete morphological characters were analysed using the Mk-model with correction for non-sampling of constant characters (88, 89), which has been well-tested (90, 91). Of the 107 discrete characters, only six were invariant (thus, invariant characters were essentially unsampled). Polymorphic morphological data (0&1) could be treated exactly as coded (e.g., 0 or 1, but *not* 2), not as total uncertainty (0 or 1 or 2): an improvement over earlier versions of BEAST. Two multistate characters (numbers 45, 47) each formed a clear morphocline and were treated as ordered. Among-character rate variation was modelled using the gamma parameter (92). We tested both an unpartitioned model for the discrete characters, and state-partitioned models (17, 93). The preferred state-partitioned model with rates scaled by state number (see 93) fitted much better than a simple unpartitioned model (Bayes Factor ~2000), and approximately as well as a more complex model where such rates were estimated from the data (Bayes Factor difference <1).

Meristic characters were analyzed using a high-state ordered Mk-model. Essentially, meristic characters were treated as ordered multistate characters with very large numbers of states, the possible states being the minimum observed value, the maximum observed value, and all intervening whole numbers. Changes between any two states were constrained to pass through all intervening values (states).

Continuous characters were logged (to base e) to reduce heteroscedasticity; and all values had 1 added (to remove negative values). The transformed data were then analyzed using strict and relaxed Brownian motion models implemented in BEAST (94), with the average evolutionary rate (i.e., variance) for each continuous character estimated separately.

The most appropriate available tree prior in BEAST 1.8 was the birth-death serial-sampling model (BDSS: 95). However, additional tests suggested the (unavailable) sampled ancestor-birth death (SABD) model would have yielded very similar results regarding macroevolutionary dynamics. The SABD tree prior is available in BEAST2 (17), but cannot yet be implemented in connection with epoch clocks nor with continuous traits, and thus could not be used for the current full dataset. However, analysis of only the discrete traits using the BDSS and SABD

models was possible in BEAST2 under both strict and UCLN clocks; the BDSS and SABD tree priors always yielded broadly similar topologies and divergence dates.

Branch-specific rate variation for the discrete, meristic and continuous characters was modelled using three very different clock models, as outlined below:

Strict clock. This model assumes that all branches have the same evolutionary rate. The overall rate of evolution of each dataset (discrete, meristic, continuous) was estimated separately, but for each class of characters, all branches share the same rate of change. The dated tree from this model is shown below in Fig. S2.

Epoch clock. This model assumes rates vary across time slices, but rates within each time slice are the same across branches. Branches spanning multiple time slices are allocated multiple rates (i.e., the segment of branch within time slice *a* is allocated the rate for time slice *a*, etc.). Three epochs were employed, corresponding to the major subdivisions of the Cambrian Period: Terreneuvian+Series 2, Miaolingian, and Furongian – here informally termed 'early', 'middle' and 'late' Cambrian, respectively. Separate epoch clocks were used for discrete, meristic and continuous characters, meaning that evolutionary rates across the 'early', 'middle' and 'late' Cambrian were estimated independently for each class of characters. This allows each class of characters to independently elucidate rate changes across the three time slices. The dated tree from this clock model is shown in (main) Fig. 1, along with the estimated rates of evolution for discrete, meristic and continuous characters across the 'early', 'middle' and 'late' Cambrian. Under the epoch clock, rates of evolution for all three classes of characters are surprisingly homogeneous across the entire Cambrian (Table S1).

The main Epoch clock analyses discussed above did not have any constraints on root node age. However, the Epoch clock analyses were also repeated with a root age constrained to 522 Ma (13). For these analyses, a new Epoch (Epoch 0) had to be introduced for the period spanning 522 to 519 Ma. This was because analyses with a young root constraint necessarily result in faster rates near the base, before the oldest fossils, and these (purely hypothetical) fast rates should not be averaged into the same epoch as the rates directly inferred from the earliest fossil evidence. Constraining root age to 522 Ma did not change the results here (Table S1). As expected, Epoch 0 (before the first fossils analyzed) had elevated rates, but the rates across Epochs 1, 2 and 3, which were empirically constrained by direct fossil evidence, remained very similar to each other. Note that in the main analyses, Epoch 0 is combined with Epoch 1: Epoch 0 cannot be implemented as a separate time slice, as without the root node age constraint (and of course no tip dates), there would be no temporal constraints on this part of the tree.

Uncorrelated lognormal (UCLN) relaxed clock. This assumes rates vary independently across branches, with the values exhibiting a lognormal distribution – the variance of this distribution is estimated empirically from the data, again separately for each class of characters; the mean rate is estimated using a separate parameter and thus is fixed to 1 to avoid redundancy. Again, rate variation across the tree (and thus through time) was estimated independently for each class of characters. The dated tree from this clock model is shown in (main) Fig. 2, along with the estimated rates of discrete, meristic and continuous characters plotted against branch age. Under the UCLN relaxed clock, rates of evolution for all three classes of characters are again surprisingly homogeneous across the entire Cambrian (Fig. 3).

The fit of these different clock models was evaluated using Bayes Factors (*sensu* [96]), calculated using stepping-stone sampling (97). As might be expected, given the homogeneity of rates across time, the strict clock fitted the data substantially *better* than the more complex

epoch-clock model (BF ~76). Bayes Factors could not be accurately calculated for the UCLN clock model, as stepping-stone analyses aborted mid-way despite repeated attempts.

2b. Parsimony analysis

Parsimony analyses used TNT v1.5 (98). As before, all discrete characters were unordered except for characters 45 and 47. Continuous characters were treated exactly as such, and were rescaled so that the maximum difference (lowest to highest value) spanned 0 to 2, i.e., continuous traits were rescaled to be equivalent to a 3-state discrete character. This rescaling value was chosen because the mean number of states for the 107 discrete characters was ~3 (2.65). Meristic characters were also rescaled to be equivalent to ordered discrete 3-state characters, for the same reasons.

The discrete, (scaled) meristic, and (scaled) continuous data were analyzed simultaneously using the TBR search algorithm, with 10000 replicates each holding up to 1000000 trees. Two searches were performed under these settings and both gave the same best single tree, with the same length (622.744) and topology (shown in Fig. S3, rooted to match Figs. 1 and 2). 100 simple (unpartitioned) bootstrap replicates were performed, with parsimony-uninformative characters excluded during bootstrapping; values shown in Fig. S3. Files for the TNT analyses are available in Dryad Dataset 5.

2c. Implications for trilobite systematics

Despite the considerable phylogenetic uncertainty across analyses, including the variable position of certain taxa/clades (Figs. 1, 2, S2, S3), our results are integrated across the full suite of sampled trees. Thus, the inferred rates of morphological evolution for Cambrian trilobites are robust, regardless of differing tree topologies.

Our results reveal only a very limited number of stable clades that are retrieved across all Bayesian and parsimony analyses (Figs. 1, 2, S2, S3). These include: the Eodiscida (*sensu* Adrain [18]); the Fallotaspididae (*Fallotaspis* + *Choubertella* + *Daguinaspis*); a larger monophyletic grouping of olenellines (Olenellidae + Biceratopsidae + Nevadiidae + Holmiidae + Judomiidae); and several redlichiine clades, such as Redlichiidae + Yinitidae, Bathynotidae, Emuelloidea (Emuellidae + Megapharanaspidae), and Paradoxidoidea (Paradoxididae + Xystriduridae). With the exception of the Eodiscida, all of these clades are traditionally placed in the order Redlichiida (e.g. 18, 41, 99). However, relationships between and within its major subdivisions (i.e., the suborders Redlichiina and Olenellina, and associated superfamilies) are often considered to be polyphyletic or paraphyletic (15, 57, 60, 86, 87, 100-102). In reference to recent classification schemes (e.g. 18, 41), our results show that Redlichiida is indeed non-monophyletic.

Bayesian and parsimony analyses recover a monophyletic Eodiscida (*sensu* Adrain [18]), but its position is highly labile (Figs. 1, 2, S2, S3). Parsimony retrieves a deeply nested Eodiscida that is sister to an unusual clade comprising *Akoldinioidia* (Shumardiidae), *Cedaria* (Cedariidae), *Golasaphus* (Asaphidae), *Haniwa* (Remopleurididae), and *Plethopeltis* (Plethopeltidae) (Fig. S3). Conversely, the Bayesian analyses produce relatively basal positions for Eodiscida, the tip dates forcing them stemward. Results of the multi- and single-epoch clock analyses (Figs. 1, S2) conform somewhat to previous phylogenetic hypotheses where the Eodiscida is basal to redlichiines and more derived taxa (text-fig. 14 in 61; fig. 1 in 46). The relaxed (uncorrelated lognormal) clock analysis (Fig. 2) places the Eodiscida among redlichiines, providing some

credence to Jell's (102, 103) idea that the group evolved from either redlichioids or bigotinids by paedomorphosis.

Since many higher-level trilobite groups (e.g., orders, suborders and superfamilies) were largely devised prior to the application of parsimony- and Bayesian-based methods used in phylogenetic analyses, it is unsurprising that our results do not conform to conventional (41, 99) or more recent (alternative) classification schemes (18). Although the order Ptychopariida has long been considered a 'wastebasket' taxon for a huge number of trilobites and is clearly polyphyletic (18, 37), our results also challenge the monophyly of other traditional groups, for example, the order Corynexochida. In particular, members of the suborder Corynexochina, which are typified by the fusion of the hypostome and rostral plate (41, 58, 61, 62, 99; *Zacanthopsis palmeri* is an exception [104]), are not monophyletic in our Bayesian or parsimony analyses. In fact, our results show that this character is homoplasious, as it also occurs in the Bathynotidae – a family closely allied with the Emuelloidea (order Redlichiida) (Figs. 1, S2, S3), primarily based on cephalic features and trunk tagmosis. Some other traits previously considered diagnostic (synapomorphic) of large clades, such as the ventral median suture of the order Asaphida (58, 61, 63), are also homoplasious across the Bayesian and parsimony trees, as suspected in other studies (65-68, 105-107). While the purpose of our study is not to revise or dismantle pre-existing classification schemes, it will hopefully provide a robust phylogenetic framework for future work on trilobite systematics.

References (1–36 in main text)

37. Lieberman BS, Karim TS (2010) Tracing the trilobite tree from the root to the tips: A model marriage of fossils and phylogeny. *Arth Struct Dev* 39:111–123.
38. Lee MSY, Palci A (2015) Morphological phylogenetics in the genomic age. *Curr Biol* 25:R922–R929.
39. Matzke NJ, Irmis RB (2018) Including autapomorphies is important for paleontological tip-dating with clocklike data, but not with non-clock data. *PeerJ* 6:e4553.
40. Jell PA, Adrain JM (2003) Available generic names for trilobites. *Mem Qld Mus* 48:331–553.
41. Fortey RA (1997) Classification. *Treatise on Invertebrate Paleontology, Part O, Arthropoda 1, Trilobita (Revised)*, ed Kaesler RL (Geological Society of America and University of Kansas Press), pp 289–302.
42. Müller KJ, Walossek D (1987) Morphology, ontogeny, and life habit of *Agnostus pisiformis* from the Upper Cambrian of Sweden. *Fossils and Strata* 19:1–124.
43. Robison RA (1972) Hypostoma of agnostid trilobites. *Lethaia* 5:239–248.
44. Shergold JH (1991) Protaspid and early meraspid growth stages of the eodiscoid trilobite *Pagetia ocellata* Jell, and their implications for classification. *Alcheringa* 15:65–86.
45. Walossek D, Müller KJ (1990) Upper Cambrian stem-lineage crustaceans and their bearing upon the monophyletic origin of Crustacea and the position of *Agnostus*. *Lethaia* 23:409–427.
46. Ramsköld L, Edgecombe GD (1991) Trilobite monophyly revisited. *Hist Biol* 4:267–283.
47. Bergström J, Hou X-G (2005) Early Palaeozoic non-lamellipedian arthropods. *Crustacea and Arthropod Relationships*, eds Koenemann S, Jenner RA (Taylor & Francis), pp 73–93.

48. Haug JT, Maas A, Waloszek D (2010) †*Henningsmoenicaris scutula*, †*Sandtorpia vestrogothiensis* gen. et sp. nov. and heterochronic events in early crustacean evolution. *Earth Env Sci Trans R Soc Edin* 100:311–350.
49. Ortega-Hernández J, Brena C (2012) Ancestral patterning of tergite formation in a centipede suggests derived mode of trunk segmentation in trilobites. *PLoS ONE* 7(12):e52623.
50. Jackson ISC, Budd GE (2017) Intraspecific morphological variation of *Agnostus pisiformis*, a Cambrian Series 3 trilobite-like arthropod. *Lethaia* 50:467–485.
51. Geyer G, Shergold JH (2000) The quest for internationally recognized divisions of Cambrian time. *Episodes* 23:188–195.
52. Geyer G, Peel JS (2011) The Hensen Gletscher Formation, North Greenland, and its bearing on the global Cambrian Series 2–Series 3 boundary. *Bull Geosci* 86:465–534.
53. Yuan J-L, Zhu X-J, Lin J-P, Zhu MY (2011) Tentative correlation of Cambrian Series 2 between South China and other continents. *Bull Geosci* 86:397–404.
54. Webster M (2011) Trilobite biostratigraphy and sequence stratigraphy of the upper Dyeran (traditional Laurentian "Lower Cambrian") in the southern Great Basin, U.S.A. *Cambrian Stratigraphy and Paleontology of Northern Arizona and Southern Nevada*, ed Hollingsworth JS, Sundberg FA, Foster JR (Museum of Northern Arizona Bulletin 67), pp 121–154.
55. Nielsen AT, Weidner T, Terfelt F, Høyberget M (2014) Upper Cambrian (Furongian) biostratigraphy in Scandinavia revisited: definition of superzones. *GFF* 136:193-197.
56. Sundberg FA, Geyer G, Kruse PD, McCollum LB, Pegel TV, Żylińska A, Zhuravlev AYu (2016) International correlation of the Cambrian Series 2-3, Stages 4-5 boundary interval. *Austral Palaeontol Mem* 49:83–124.
57. Paterson JR, Edgecombe GD (2006) The early Cambrian trilobite family Emuellidae Pocock, 1970: systematic position and revision of Australian species. *J Paleontol* 80:496–513.
58. Fortey RA (2001) Trilobite systematics: the last 75 years. *J Paleontol* 75:1141–1151.
59. Fortey RA, Whittington HB (1989) The Trilobita as a natural group. *Hist Biol* 2:125–138.
60. Palmer AR, Repina LN (1993) Through a glass darkly: Taxonomy, phylogeny, and biostratigraphy of the Olenellina. *Uni Kansas Paleontol Contrib, New Series* 3:1–35.
61. Fortey RA (1990) Ontogeny, hypostome attachment and trilobite classification. *Palaeontology* 33:529–576.
62. Whittington HB (2009) The Corynexochina (Trilobita): a poorly understood suborder. *J Paleontol* 83:1–8.
63. Fortey RA, Chatterton BDE (1988) Classification of the trilobite suborder Asaphina. *Palaeontology* 31:165–222.
64. Chatterton BDE, Ludvigsen R (1998) Upper Steptoean (Upper Cambrian) trilobites from the McKay Group of southeastern British Columbia, Canada. *The Paleontol Soc Mem* 49:1–43.
65. Park T, Choi DK (2009) Post-embryonic development of the Furongian (late Cambrian) trilobite *Tsinania canens*: implications for life mode and phylogeny. *Evol & Dev* 11:441–455.
66. Park T, Choi DK (2010) Ontogeny and ventral median suture of the ptychaspidid trilobite *Asioptychaspis subglobosa* (Sun, 1924) from the Furongian (Upper Cambrian) Hwajeol Formation, Korea. *J Paleontol* 84:309–320.

67. Zhu X, Hughes NC, Peng S (2010) Ventral structure and ontogeny of the late Furongian (Cambrian) trilobite *Guangxiaspis guangxiensis* Zhou, 1977 and the diphyletic origin of the median suture. *J Paleontol* 84:493–504.
68. Park T, Kihm JH, Kang I, Choi DK (2014) Ontogeny of a new species of the Cambrian Series 3 (middle Cambrian) trilobite genus *Liostracina* Monke, 1903 from North China and the taxonomic position of the superfamily Trinucleoidea. *J Paleontol* 88:395–402.
69. Whittington HB (1990) Articulation and exuviation in Cambrian trilobites. *Philos Trans R Soc B* 329:27–46.
70. Hughes NC (2007) The evolution of trilobite body patterning. *Ann Rev Earth Planet Sci* 35:401–434.
71. Laibl L, Fatka O, Budil P, Ahlberg P, Szabad M, Vokáč V, Kozák V (2015) The ontogeny of *Ellipsocephalus* (Trilobita) and systematic position of Ellipsocephalidae. *Alcheringa* 39:477–487.
72. Whittington HB, Kelly SRA (1997) Morphological terms applied to Trilobita. *Treatise on Invertebrate Paleontology, Part O, Arthropoda 1, Trilobita (Revised)*, ed Kaesler RL (Geological Society of America and University of Kansas Press), pp 313–329.
73. Jell PA (1975) Australian Middle Cambrian eodiscoids with a review of the superfamily. *Palaeonto Abt. A* 150:1–97.
74. Cotton TJ, Fortey RA (2005) Comparative morphology and relationships of the Agnostida. *Crustacea and Arthropod Relationships*, eds Koenemann S, Jenner RA (Taylor & Francis), pp 95–136.
75. Whittington HB (1997) Morphology of the exoskeleton. *Treatise on Invertebrate Paleontology, Part O, Arthropoda 1, Trilobita (Revised)*, ed Kaesler RL (Geological Society of America and University of Kansas Press), pp 1–85.
76. Cotton TJ (2001) The phylogeny and systematics of blind Cambrian ptychoparioid trilobites. *Palaeontology* 44:167–207.
77. Palmer AR (1998) Terminal early Cambrian extinction of the Olenellina: documentation from the Pioche Formation, Nevada. *J Paleontol* 72:650–672.
78. Cederström P, Ahlberg P, Clarkson ENK, Nilsson CH, Axheimer N (2009) The lower Cambrian eodiscoid trilobite *Calodiscus lobatus* from Sweden: morphology, ontogeny and distribution. *Palaeontology* 52:491–539.
79. Hughes NC, Minelli A, Fusco G (2006) The ontogeny of trilobite segmentation: a comparative approach. *Paleobiology* 32:602–627.
80. Drummond AJ, Suchard MA (2010) Bayesian random local clocks, or one rate to rule them all. *BMC Biology* 8:114.
81. Ronquist F, Klopfstein S, Vilhelmsen L, Schulmeister S, Murray DL, Rasnitsyn AP (2012) A total-evidence approach to dating with fossils, applied to the early radiation of the Hymenoptera. *Syst Biol* 61:973–999.
82. Bouckaert R, Heled J, Kühnert D, Vaughan T, Wu C-H, Xie D, Suchard MA, Rambaut A, Drummond AJ (2014) BEAST 2: A software platform for Bayesian evolutionary analysis. *PLoS Comp Biol* 10:e1003537.
83. Drummond AJ, Suchard MA, Xie D, Rambaut A (2012) Bayesian phylogenetics with BEAUti and the BEAST 1.7. *Mol Biol Evol* 29:1969–1973.
84. Jombart T, et al. (2014) *OutbreakTools*: a new platform for disease outbreak analysis using the R software. *Epidemics* 7:28–34.

85. Bielejec F, Lemey P, Baele G, Rambaut A, Suchard MA (2014) Inferring heterogeneous evolutionary processes through time: from sequence substitution to phylogeography. *Syst Biol* 63:493–504.
86. Lieberman BS (1998) Cladistic analysis of the early Cambrian olenelloid trilobites. *J Paleontol* 72:59–78.
87. Lieberman, BS (2001) Phylogenetic analysis of the Olenellina Walcott, 1890 (Trilobita, Cambrian). *J Paleontol* 75:96–115.
88. Lewis PO (2001) A likelihood approach to estimating phylogeny from discrete morphological character data. *Syst Biol* 50:913–925.
89. Alekseyenko AV, Lee C, Suchard MA (2008) Wagner and Dollo: a stochastic duet by composing two parsimonious solos. *Syst Biol* 57:772–784.
90. Wright AM, Hillis DM (2014) Bayesian analysis using a simple likelihood model outperforms parsimony for estimation of phylogeny from discrete morphological data. *PLoS ONE* 9:e109210.
91. O'Reilly JE, Puttick MN, Parry L, Tanner AR, Tarver JE, Fleming J, Pisani D, Donoghue PCJ (2016) Bayesian methods outperform parsimony but at the expense of precision in the estimation of phylogeny from discrete morphological data. *Biol Letters* 12:20160081.
92. Yang Z (1994) Maximum likelihood phylogenetic estimation from DNA sequences with variable rates over sites: Approximate methods. *J Mol Evol* 39:306–314.
93. King B, Qiao T, Lee MSY, Zhu M, Long JA (2017) Bayesian morphological clock methods resurrect placoderm monophyly and reveal rapid early evolution in jawed vertebrates. *Syst Biol* 66:499–516.
94. Lemey P, Rambaut A, Welch JJ, Suchard MA (2010) Phylogeography takes a relaxed random walk in continuous space and time. *Mol Biol Evol* 27:1877–1885.
95. Stadler T, Yang Z (2013) Dating phylogenies with sequentially sampled tips. *Syst Biol* 62:674–688.
96. Kass RE, Raftery AE (1995) Bayes factors. *J Amer Stat Assoc* 90:773–795.
97. Xie W, Lewis PO, Fan Y, Kuo L, Chen M-H (2011) Improving marginal likelihood estimation for Bayesian phylogenetic model selection. *Syst Biol* 60:150–160.
98. Goloboff PA, Farris JS, Nixon KC (2008) TNT, a free program for phylogenetic analysis. *Cladistics* 24:774–786.
99. Harrington HJ, Henningsmoen G, Howell BF, Jaanusson V, Lochman-Balk C, Moore RC, Poulsen C, Rasetti F, Richter E, Richter R, Schmidt H, Sdzuy K, Struve W, Størmer L, Stubblefield CJ, Tripp R, Weller JM, Whittington HB (1959) *Treatise on Invertebrate Paleontology, Part O, Arthropoda 1* (Geological Society of America, New York and University of Kansas Press, Lawrence).
100. Geyer G (1996) The Moroccan fallotaspidid trilobites revisited. *Beringeria* 18:89–199.
101. Lieberman BS (1999) Testing the Darwinian legacy of the Cambrian radiation using trilobite phylogeny and biogeography. *J Paleontol* 73:176–181.
102. Jell PA (1997) Introduction to suborder Eodiscina. *Treatise on Invertebrate Paleontology, Part O, Arthropoda 1, Trilobita (Revised)*, ed Kaesler RL (Geological Society of America and University of Kansas Press), pp 384–386.
103. Jell PA (2003) Phylogeny of early Cambrian trilobites. *Special Papers in Palaeo* 70:45–57.
104. Hopkins MJ, Webster M (2009) Ontogeny and geographic variation of a new species of the corynexochine trilobite *Zacanthopsis* (Dyeran, Cambrian). *J Paleontol* 83:524–547.

105. Whittington HB (2003) The upper Cambrian trilobites *Plethopeltis* and *Leiocoryphe*: morphology, paedomorphism, classification. *J Paleontol* 77:698–705.
106. Zhu X, Hughes NC, Peng S (2007) On a new species of *Shergoldia* Zhang & Jell, 1987 (Trilobita), the family Tsinaniidae and the order Asaphida. *AAP Memoirs* 34:243–253.
107. Park T, Choi DK (2011) Ontogeny of the Furongian (late Cambrian) remopleuridioid trilobite *Haniwa quadrata* Kobayashi, 1933 from Korea: implications for trilobite taxonomy. *Geol Mag* 148:288–303.

Table S1. Absolute (upper panel) and scaled (lower panel) evolutionary rates across the 'early' (Series 2 = Epoch 1), 'middle' (Miaolingian = Epoch 2), and 'late' (Furongian = Epoch 3) Cambrian. Scaled rates are simply the absolute rates uniformly adjusted so that the rate in the middle Cambrian (Epoch 2) is set to 1. Rates were calculated separately for discrete, meristic and continuous characters, with either no node age constraints (upper value in each cell), or the root age constrained to be 522 Ma (lower value). In the 522 Ma analysis, a new time slice (Epoch 0) had to be introduced to avoid conflating purely hypothetical fast rates in the earliest portion of the tree (before the first fossils) with rates in Epoch 1, which are directly inferred from fossil evidence (explained in Materials and Methods: Section 2a). Absolute rates are the average instantaneous rate of change across all characters of a given type (per lineage per Ma) in a particular epoch. For discrete and meristic traits, absolute rates equal the estimated rate parameter in the substitution matrix x equilibrium base frequencies x epoch rate / any relevant n-state rate scalar. For continuous traits, absolute rates equate to the average variance in the Brownian motion model across all 6 continuous traits x epoch rate / combined continuous rate scalar.

ABSOLUTE RATES

		Epoch 0	Epoch 1	Epoch 2	Epoch 3
Discrete	Root unconst	*	0.0065	0.0052	0.0048
	Root 522Ma	0.0349	0.0074	0.0062	0.0061
Meristic	Root unconst	*	0.0380	0.0320	0.0418
	Root 522Ma	0.6208	0.0106	0.0513	0.0525
Continuous	Root unconst	*	0.0334	0.0313	0.0256
	Root 522Ma	0.0894	0.0420	0.0443	0.0338

SCALED RATES

		Epoch 0	Epoch 1	Epoch 2	Epoch 3
Discrete	Root unconst	*	1.24	1.00	0.92
	Root 522Ma	5.62	1.19	1.00	0.98
Meristic	Root unconst	*	1.19	1.00	1.31
	Root 522Ma	12.10	0.21	1.00	1.02
Continuous	Root unconst	*	1.07	1.00	0.82
	Root 522Ma	2.02	0.95	1.00	0.76

* = Needs to be combined with Epoch 1 if root age is unconstrained (see SI text)

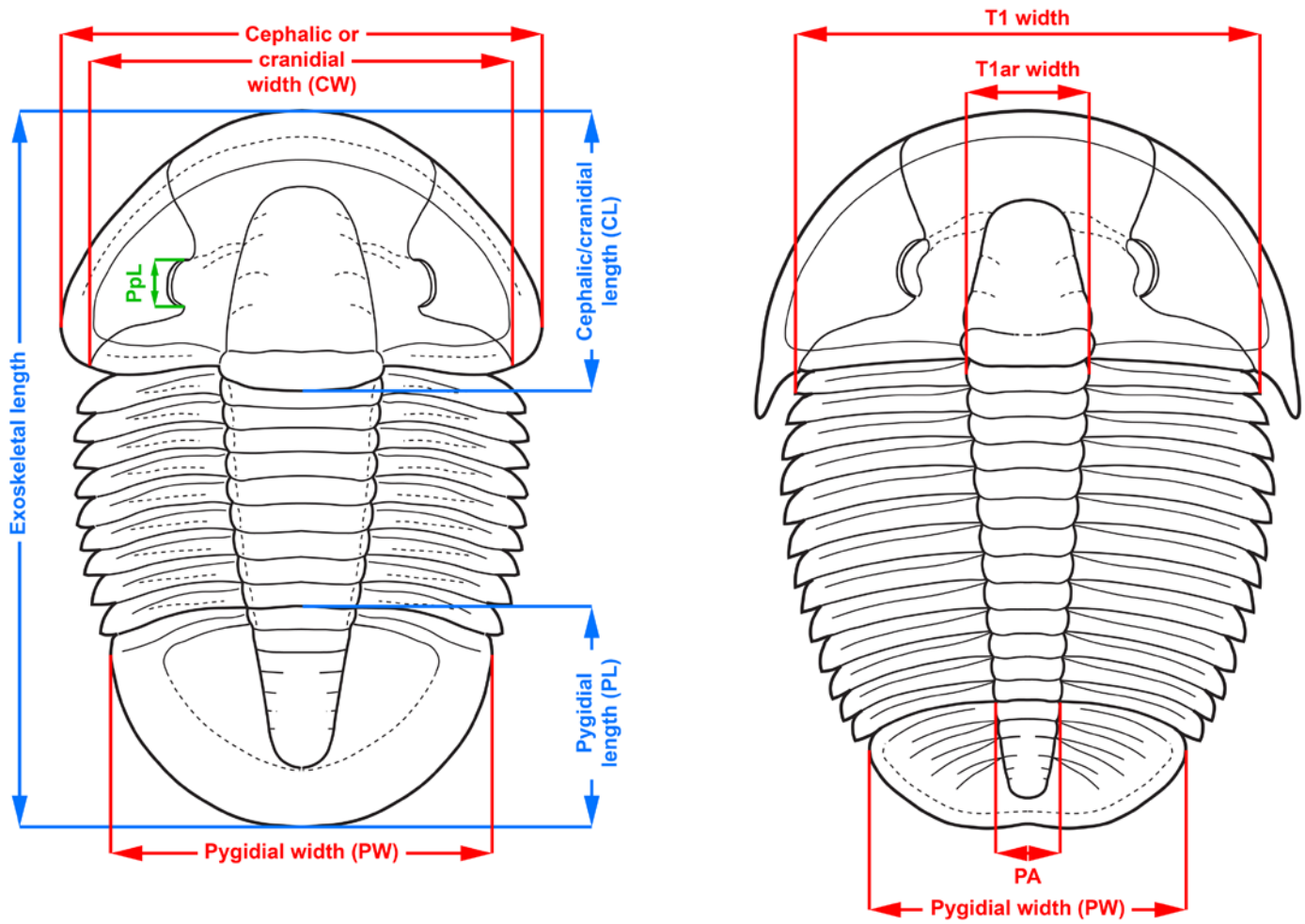


Fig. S1. Measurements for continuous characters. Character 110: Maximum sagittal exoskeletal length (in millimetres). Character 111: Cephalic/cranidial:pygidial area ratio = (CLxCW) ÷ (PLxPW). CL excludes occipital spine. For isolated cranidia, CW represents the max. transverse width, typically between distal tips of the posterolateral projections or the palpebral lobes. In cases where librigenae are also known (and represent a size [incl. facial suture line] match for the cranidium), their width is added to produce an approximate cephalic width. This measurement also excludes genal spines. PL excludes the articulating half ring and the pygidial spines. PW excludes pygidial spines. Character 112: Ratio of thoracic:pygidial segments in trunk is calculated using data from meristic characters 108 and 109. Character 113: Length of palpebral lobe from γ to ε (as % of sag. cephalic/cranidial length) = PpL ÷ CL. Character 114: Width (tr.) of thoracic axial ring on T1 (as % of the entire width of T1) = T1ar ÷ T1. Character 115: Maximum width (tr.) of pygidial axis (as % of max. pygidial width) = PA ÷ PW. Line drawings provided by Sam Gon III (www.trilobites.info).

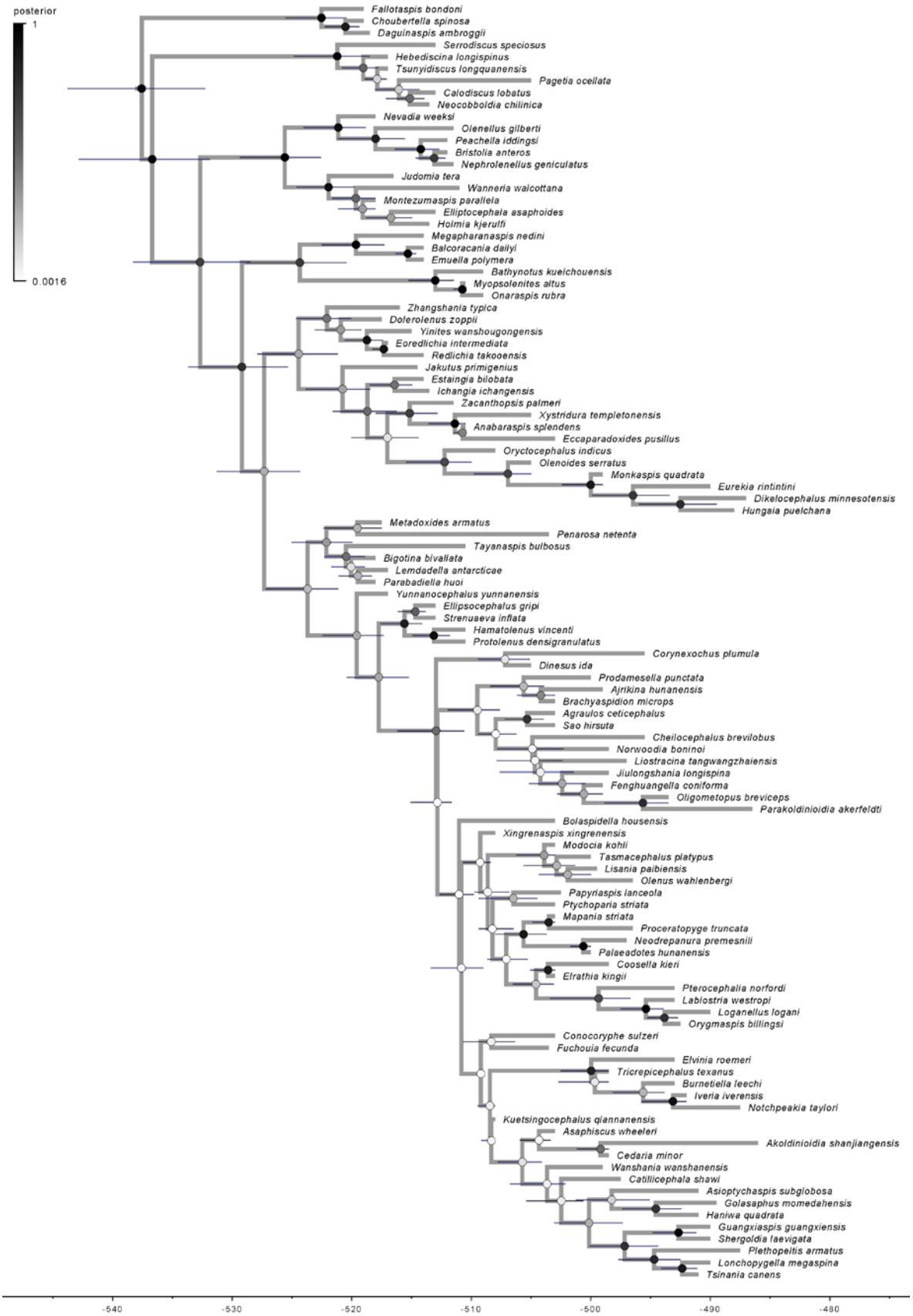


Fig. S2. Dated time-tree of Cambrian trilobites inferred from tip-dated Bayesian analyses of discrete, meristic and continuous traits under a strict clock, which assumes the rate of evolution for each type of character is uniform across all branches; this rate was estimated independently for each type of trait. Node bars indicate 95% highest posterior density intervals for divergence dates, node circles indicate posterior support for clade.

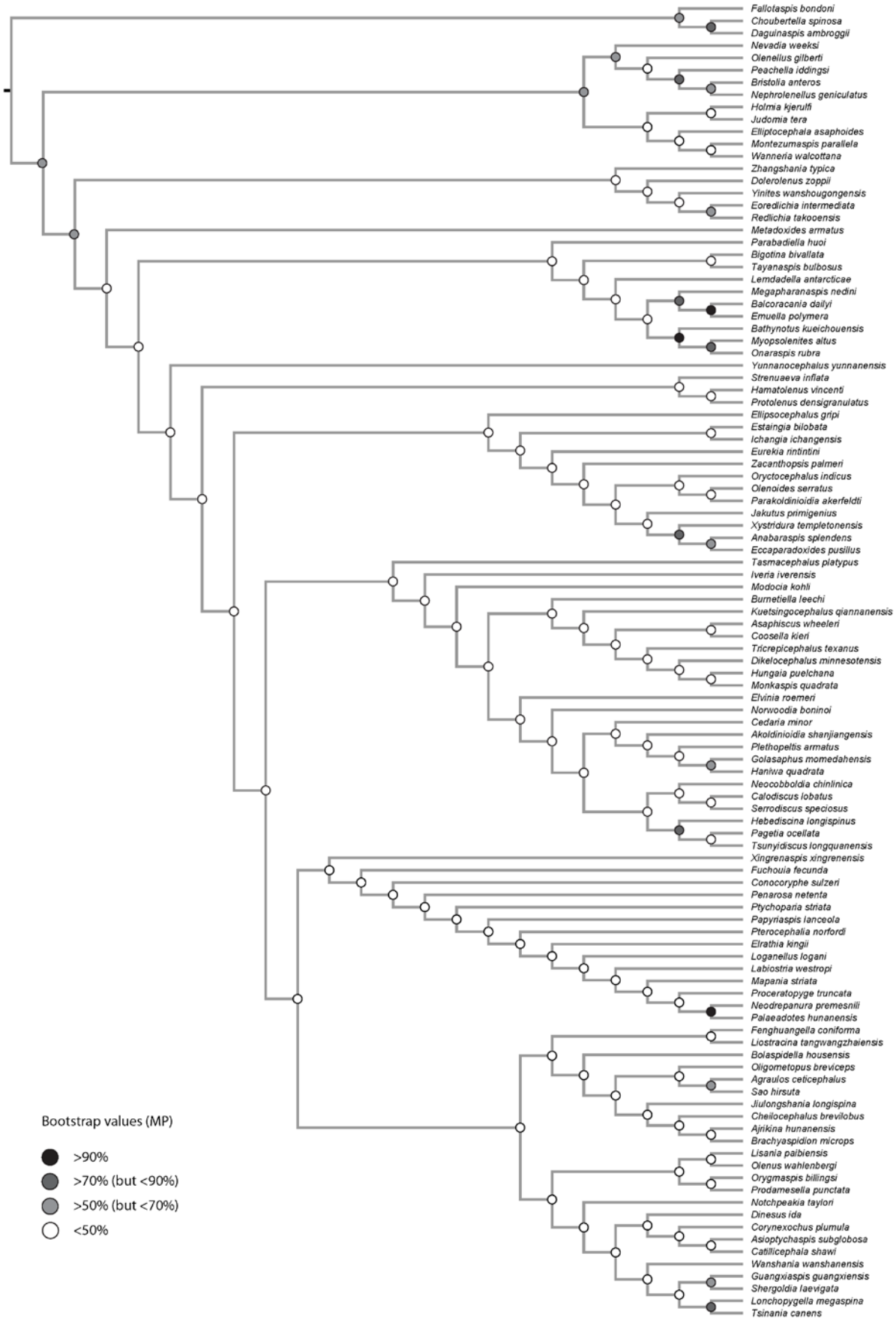


Fig. S3. Parsimony tree based on combined analysis of discrete, meristic and continuous characters. Single most-parsimonious tree of length 622.744; circles at nodes denote bootstrap values.